CardamomOT — analysis report

/Users/eventre/Documents/GitHub/CardamomOT/experimental_datasets/Semrau

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Run

Split	train
Stimulus penalisation	1.0
Prior penalisation	1.0
Genes	41
Stimuli	1
Observed cells	900
Timepoints	0, 6, 12, 24, 36, 48, 60, 72, 96
Cell types	Pluripotency, Neuroectoderm, Extraembryonic_Endoderm, Post-implantation_Epiblast
Network shown	inter_simul.npy
Embeddings	UMAP; mRNA: log1p=True, normalise=False, cell_depth_for_representation=True (observed / depth factor, model draws at the reference depth)

Perturbations (perturbation_simulation)

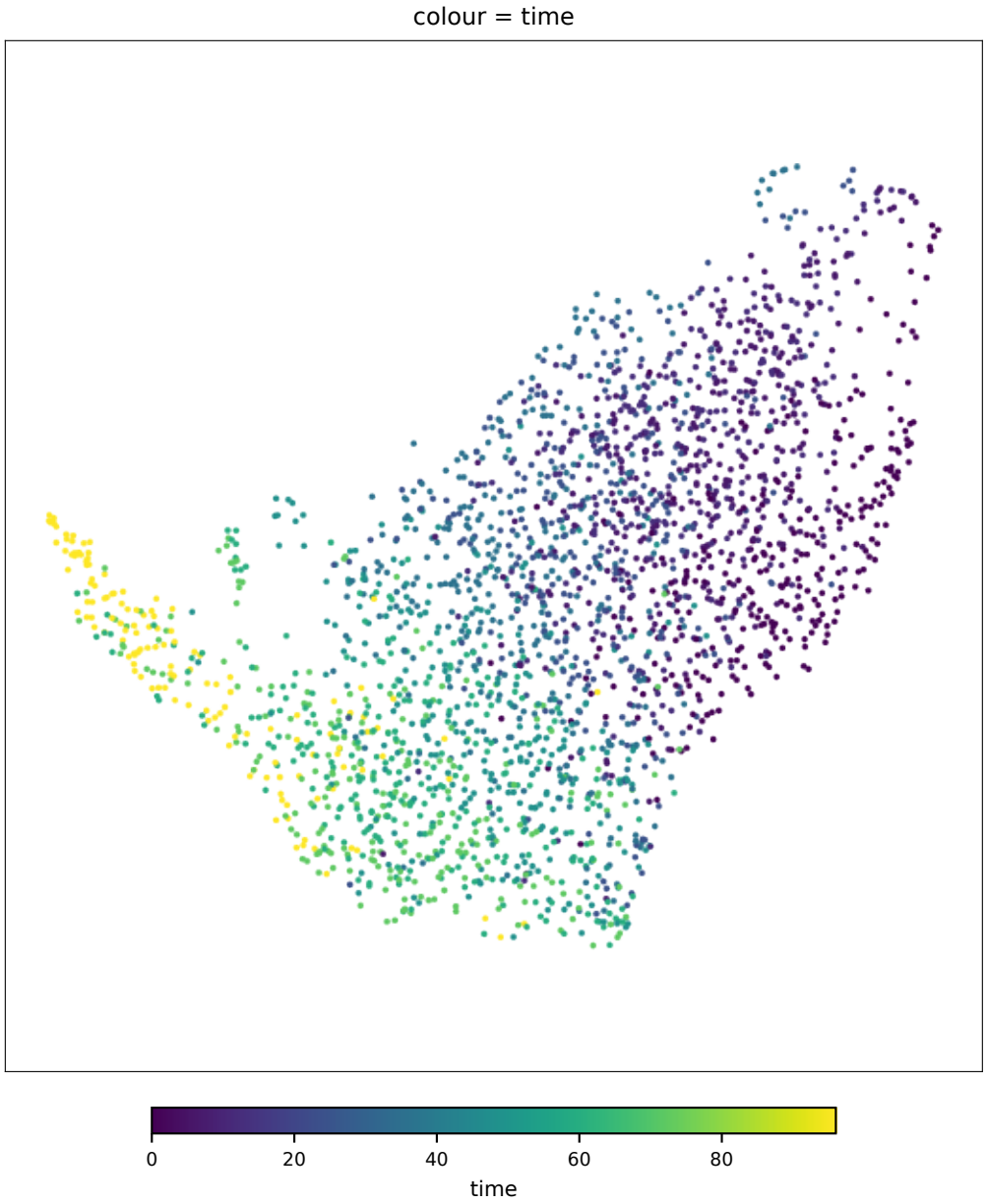
- ✓ P1 KO Dnmt3a
- ✓ P2 OV Dnmt3a

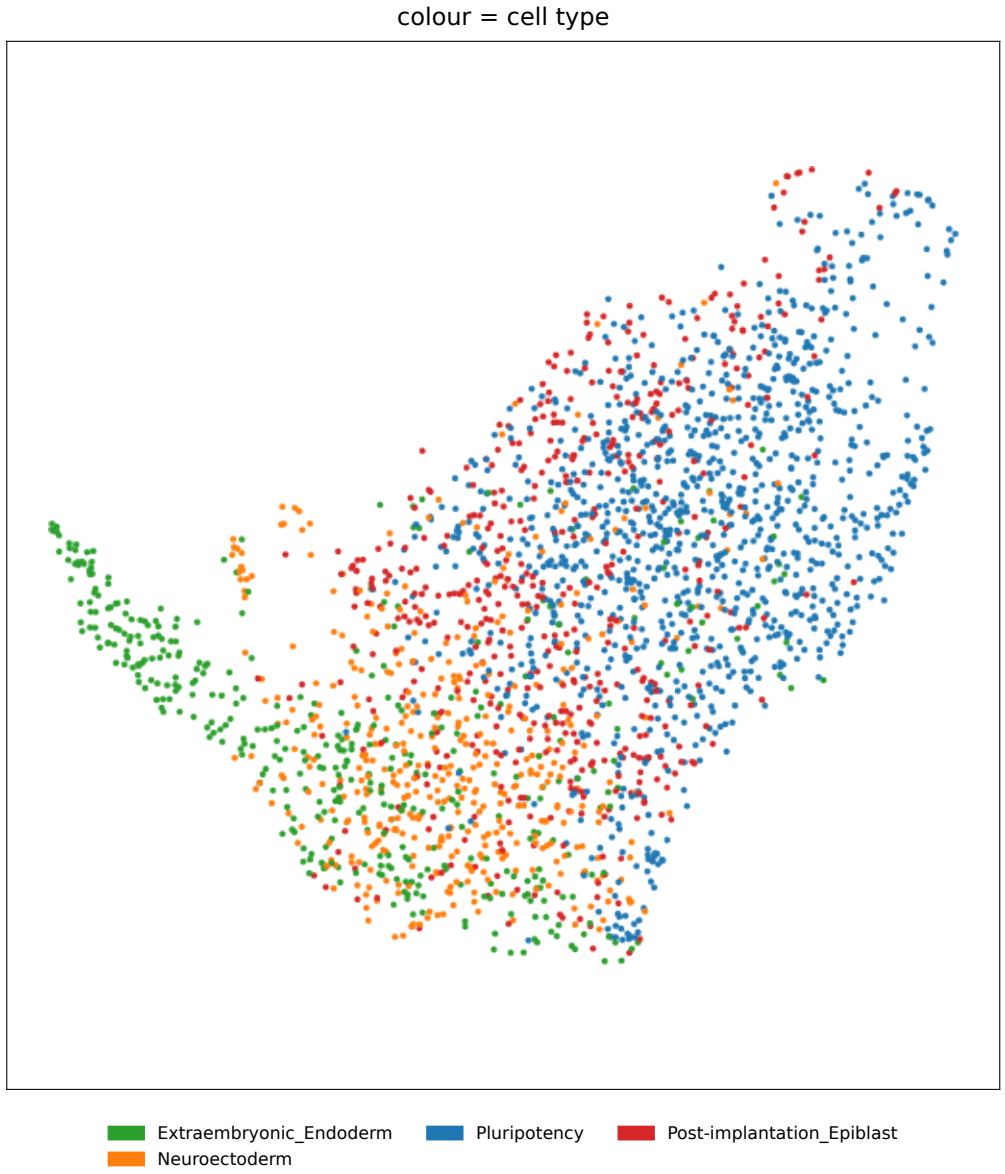
Contents

Data — UMAP of all the cells of Data/data.h5ad (time, samples); teaser: classical OT vs CardamomOT displacement fields (all times, then per interval), then cell-type transitio
1. Generative model — UMAPs of data, trajectories, NB mixture, network modes and simulation; cell-type proportions; gene-pair correlations; proteins
2. Gene regulatory network — regulatory power (violin plots), top-20 regulators + stimulus and top-20 regulated genes (mutual top-10 edges)
3. In-silico perturbations — overview across KO/OV, then one page per perturbation
4. Proliferation — prior vs learned net rates, population growth, proteins driving growth; protein dilution at the birth rate
5. Learned dynamics — mRNA and protein fields (mechanistic velocity, displacement along trajectories), summary on mRNA
6. Held-out test cells — predictions with the network fixed vs the test data

Data — UMAP of all the cells

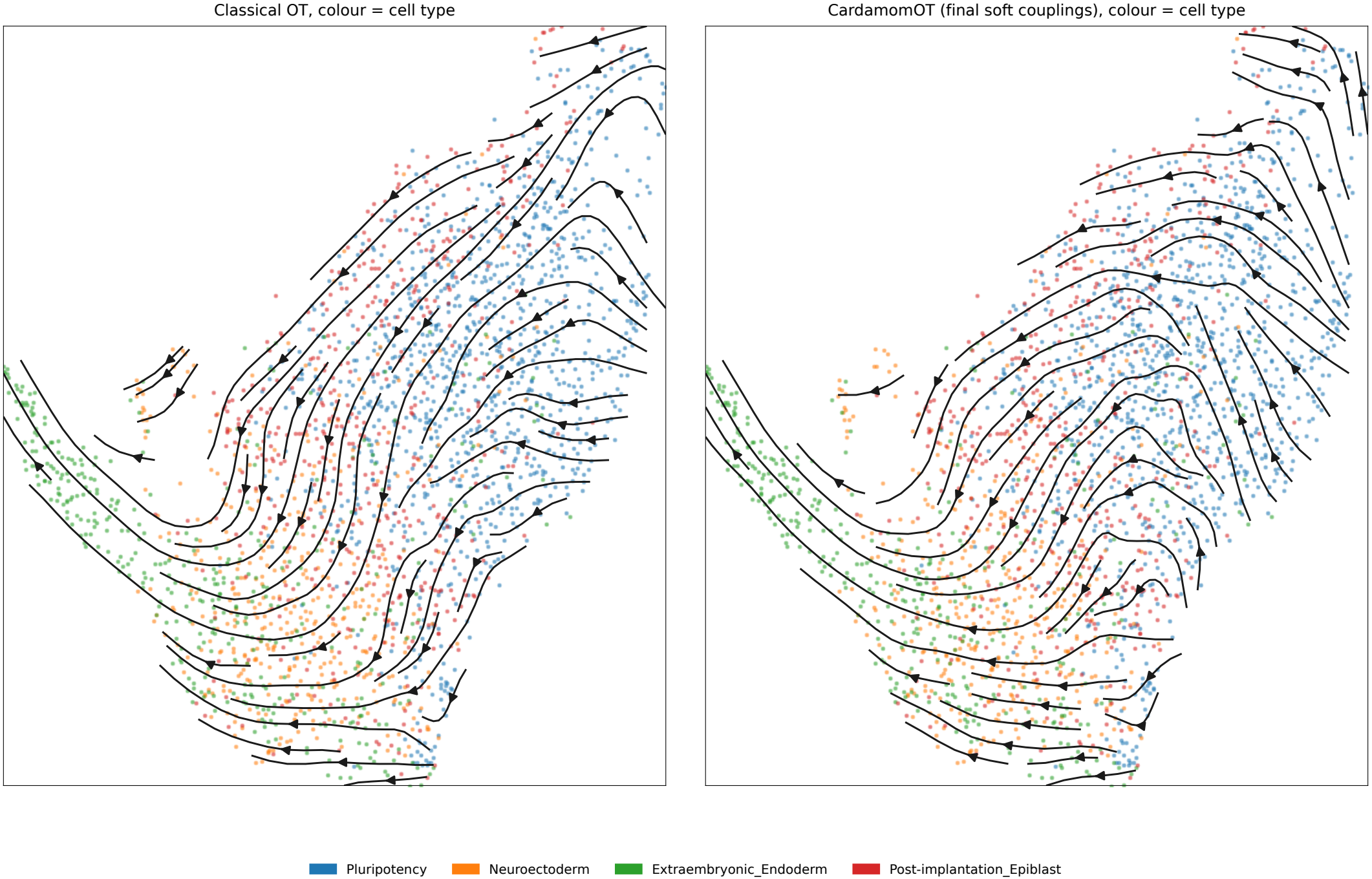
Data/data.h5ad, 2449 cells: log1p(counts), 41 highly variable genes, PCA (40 components), kNN graph, UMAP. Stored in cardamomOT/embedding_data.npz.





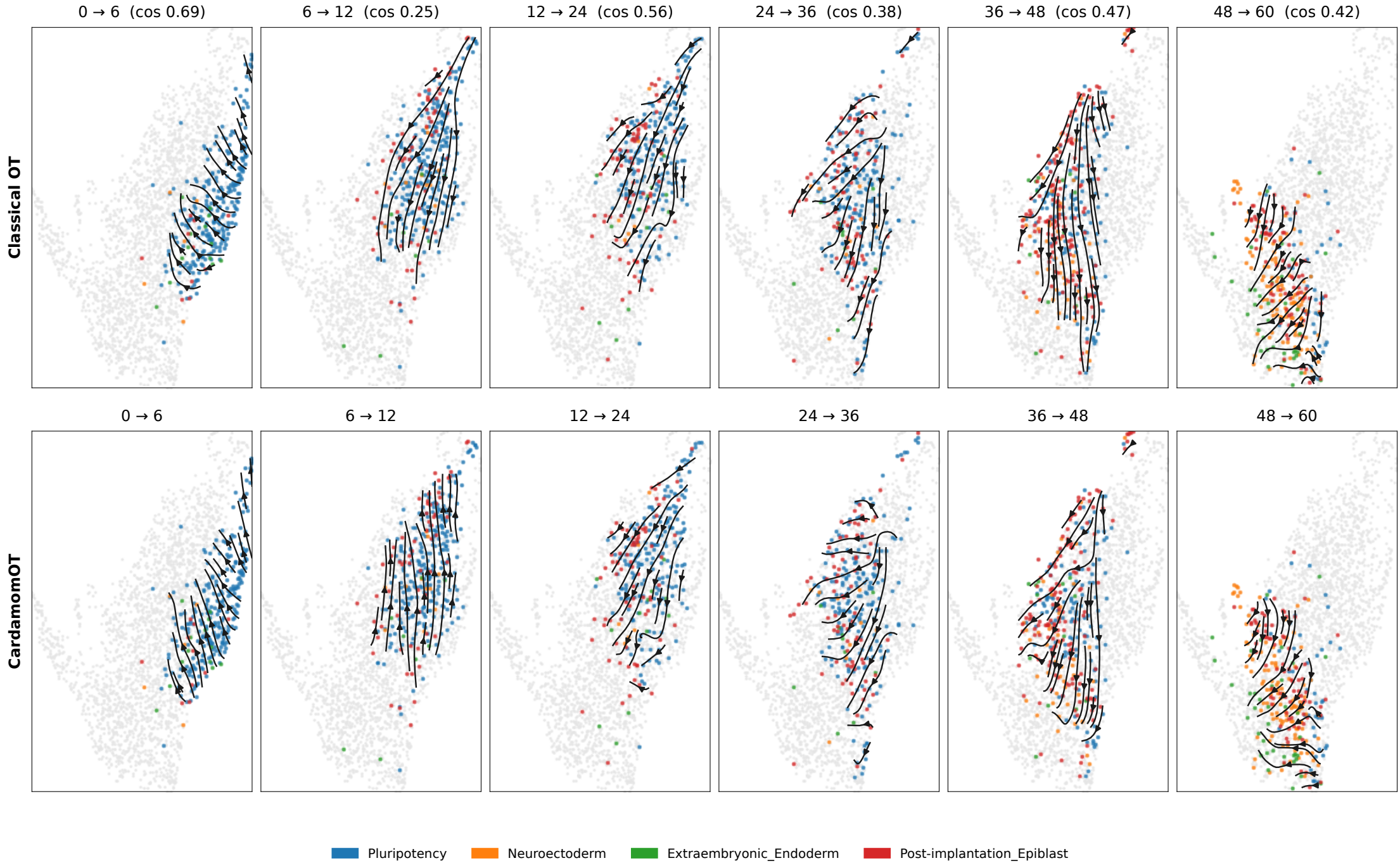
Teaser — displacement fields of classical OT and CardamomOT

Displacement to the mean descendant (x(t+1) − x(t), not divided by Δt), all times. Agreement on the cells with both direct displacements: weighted cosine 0.58 (PCA space). CardamomOT: final soft couplings of the trajectories (900 cells); cells never reached or outside the training set: Gaussian kernel on the model mRNAs.



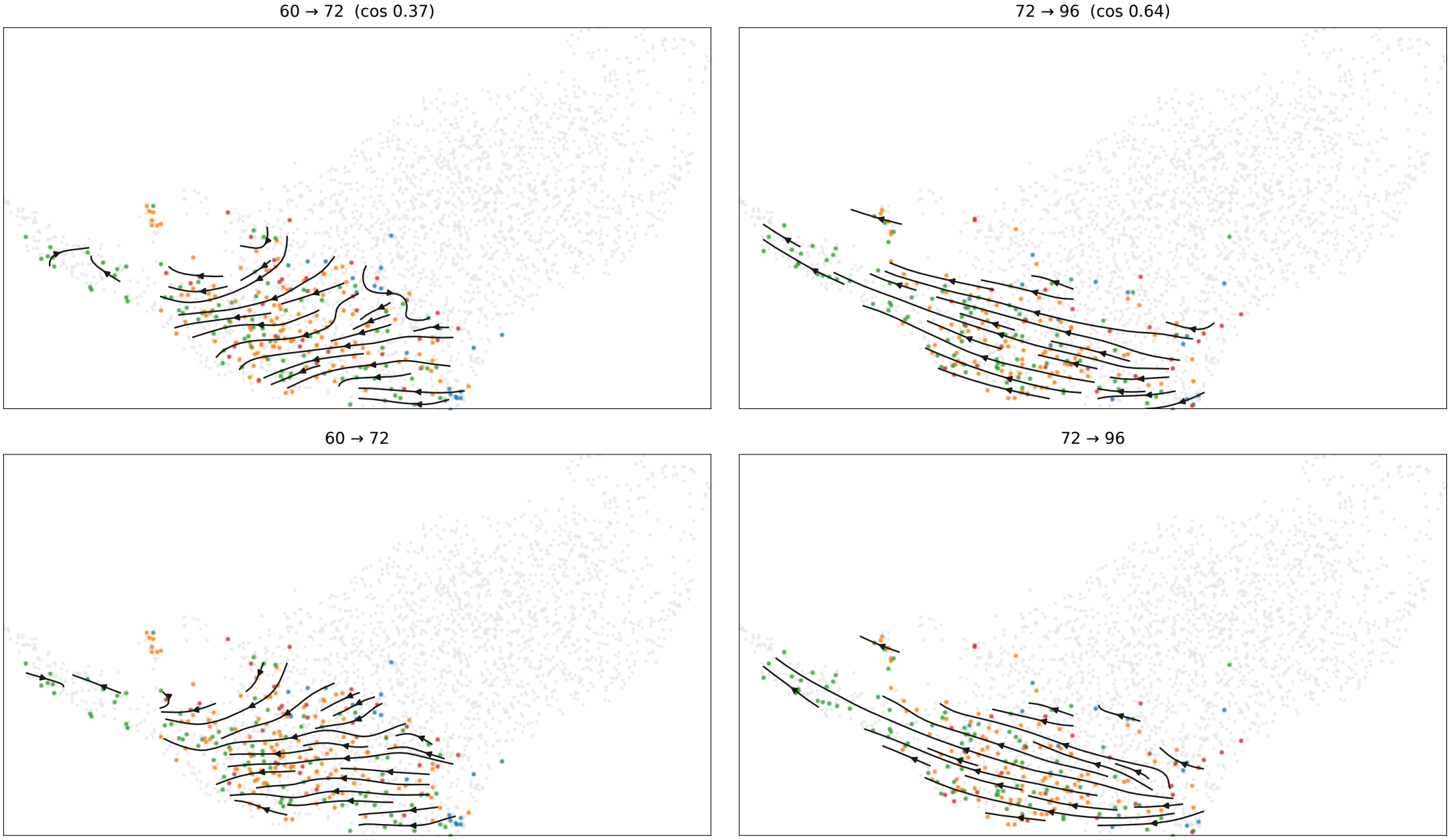
Teaser — displacement fields over time, classical OT vs CardamomOT

Cells of each t_from (coloured; other cells in grey) towards their mean descendant at t_to, kNN projection among the cells of t_from and t_to. cos: weighted cosine between the two methods on the cells of t_from with both direct displacements (PCA space).



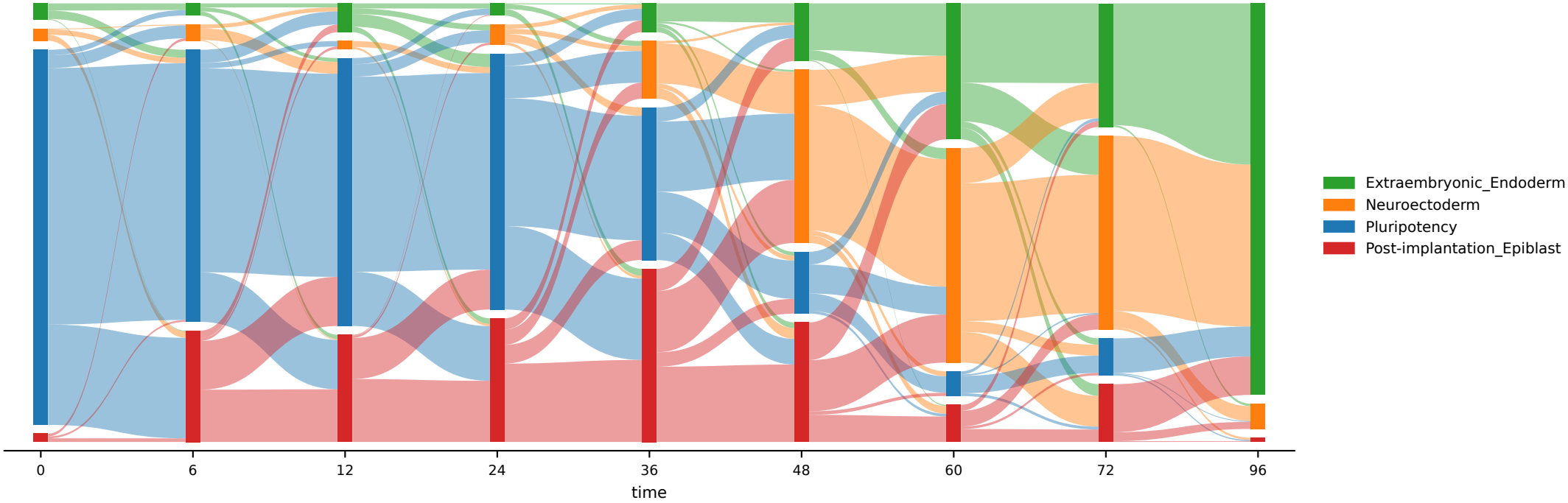
Teaser — displacement fields over time, classical OT vs CardamomOT

Cells of each t_from (coloured; other cells in grey) towards their mean descendant at t_to, kNN projection among the cells of t_from and t_to. cos: weighted cosine between the two methods on the cells of t_from with both direct displacements (PCA space).



Classical OT — cell-type transitions and fate genes

Waddington-OT couplings (every gene, train cells, per sample, growth from the net rate), pooled over the samples. Sankey: node = share of the cell type; links leave ∝ fate and arrive ∝ ancestor probabilities. Table: weighted Welch t, genes of both top 10 in red (classical_OT/fate_genes.csv).

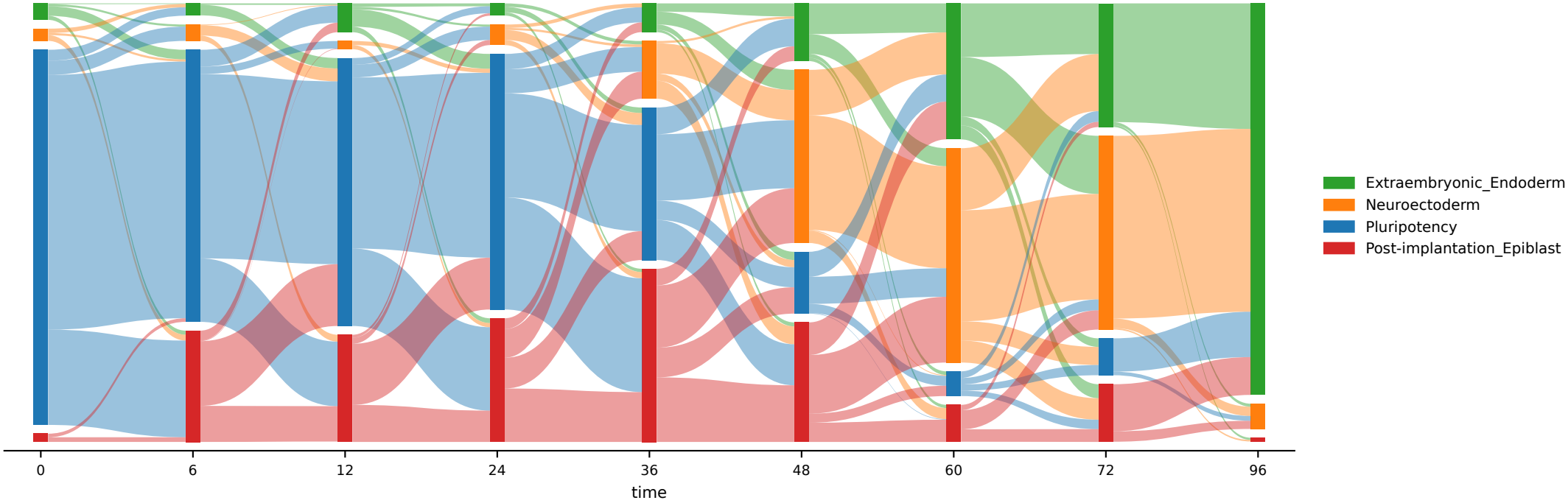


	Top-10 genes up in the ancestors (vs ancestors of the other cells)									
Extraembryonic_Endoderm	**Sparc**	**Lamb1**	**Col4a2**	Cd24a	**Dab2**	Cdh2	**Gata6**	**Krt8**	Sox11	**Pdgfra**
Neuroectoderm	**Hoxb2**	**Hoxa1**	**Sox1**	**Meis1**	**Sox11**	**Cd24a**	**Cdh2**	**Pax6**	**Nes**	Pou3f1
Pluripotency	**Zfp42**	**Esrrb**	**Klf2**	**Pou5f1**	**Sox2**	**Fgf4**	**Klf4**	**Jarid2**	**Nr0b1**	**Dppa3**
Post-implantation_Epiblast	**Dppa2**	**Dnmt3a**	**Zic2**	**Dnmt3b**	**Zfp42**	**Dppa4**	**Zic3**	Klf2	Esrrb	Klf4

	Top-10 genes up in the cell type (vs the other cells)									
	Sparc	**Lamb1**	**Col4a2**	**Dab2**	**Gata6**	Fst	**Krt8**	**Pdgfra**	Gata4	Sox17
	Cd24a	**Hoxb2**	**Cdh2**	**Sox1**	**Meis1**	**Sox11**	**Pax6**	**Hoxa1**	**Nes**	Sox9
	Zfp42	**Esrrb**	**Pou5f1**	**Klf2**	**Sox2**	**Fgf4**	**Jarid2**	**Klf4**	**Nr0b1**	**Dppa3**
	Dppa2	**Dnmt3a**	**Zic2**	**Dppa4**	**Dnmt3b**	**Zic3**	Pou3f1	Cer1	Wnt8a	**Zfp42**

Teaser — CardamomOT: cell-type transitions and fate genes

Final soft couplings of the CardamomOT trajectories (model genes, data_<split> cells), pooled over the samples. Sankey: node = share of the cell type; links leave ∝ fate and arrive ∝ ancestor probabilities. Table: weighted Welch t, genes of both top 10 in red (cardamomOT/teaser_fate_genes.csv).



	Top-10 genes up in the ancestors (vs ancestors of the other cells)	Top-10 genes up in the cell type (vs the other cells)
Extraembryonic_Endoderm	**Lamb1** Cdh2 **Col4a2** Meis1 **Dab2** Hoxb2 Cd24a **Sparc** **Gata6** Pax6	**Sparc** **Lamb1** **Col4a2** **Dab2** **Gata6** Krt8 Fst Pdgfra Sox17 Gata4
Neuroectoderm	**Hoxb2** **Sox1** **Hoxa1** Dnmt3a **Sox11** Pou3f1 Lamb1 **Meis1** Krt8 Dab2	**Hoxb2** Cdh2 **Sox1** Cd24a **Meis1** **Sox11** Pax6 **Hoxa1** Nes Sox9
Pluripotency	**Zfp42** **Esrrb** **Klf2** **Pou5f1** **Fgf4** **Klf4** **Sox2** **Nr0b1** **Jarid2** Zic3	**Zfp42** **Pou5f1** **Esrrb** **Klf2** **Fgf4** **Sox2** **Jarid2** **Nr0b1** **Klf4** Dppa3
Post-implantation_Epiblast	**Dppa2** Pou5f1 **Dppa4** **Dppa2** Klf2 Sox2 Nr0b1 Esrrb **Zic3** **Dnmt3b**	**Dppa2** Dnmt3a **Dppa4** Zic2 **Dnmt3b** **Zic3** Pou3f1 Wnt8a Cer1 **Zfp42**

Genes of the model (41)

Data/data_train.h5ad, alphabetical order.

Ascl1
Cd24a
Cdh2
Cer1
Col4a2
Dab2
Dnmt3a
Dnmt3b
Dppa2
Dppa3
Dppa4
Esrrb
Fgf4
Fgf5
Fst
Gata4
Gata6
Hoxa1
Hoxb2
Jarid2
Klf2
Klf4
Krt8
Lamb1
Meis1
Nes
Nr0b1
Pax6
Pdgfra
Pou3f1
Pou5f1
Sox1
Sox11
Sox17
Sox2
Sox9
Sparc
Wnt8a
Zfp42
Zic2
Zic3

0. Per-cell sequencing depth

Diagnostic on the whole transcriptome within homogeneous groups (sample, time, cell type); counts are modelled as NB(k, c / s_i) when the depth factor s_i is applied.

Depth

not estimated

Reason

41 genes < 10000

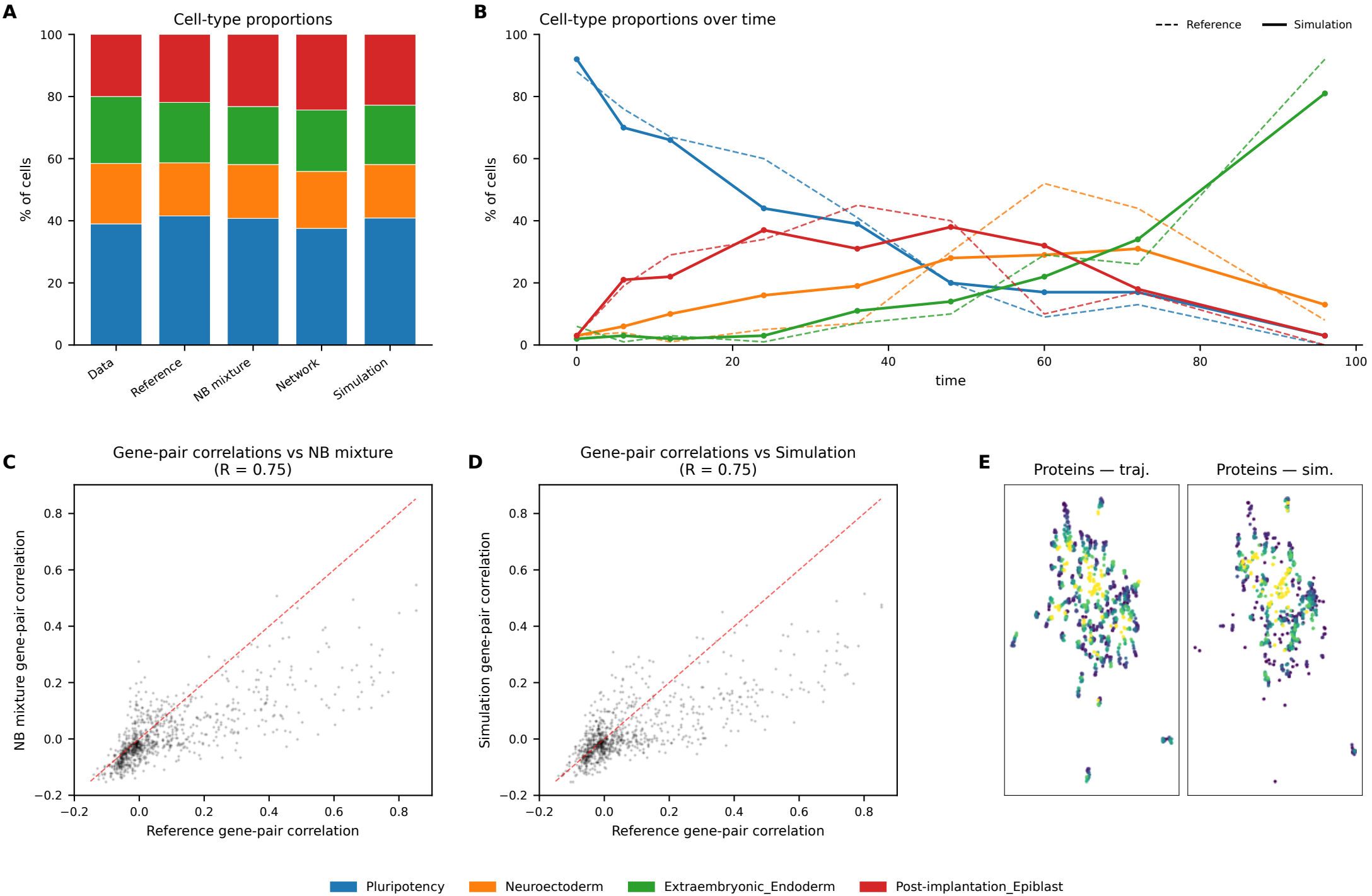
1. Generative model — trajectories and simulation

UMAP learned on the observed data; the trajectories (Reference: one state per ancestor, i.e. growth-weighted by exp ∫R_opt, as the simulation has proliferation), the NB mixture and network-driven modes along them and the full simulation are projected onto it.



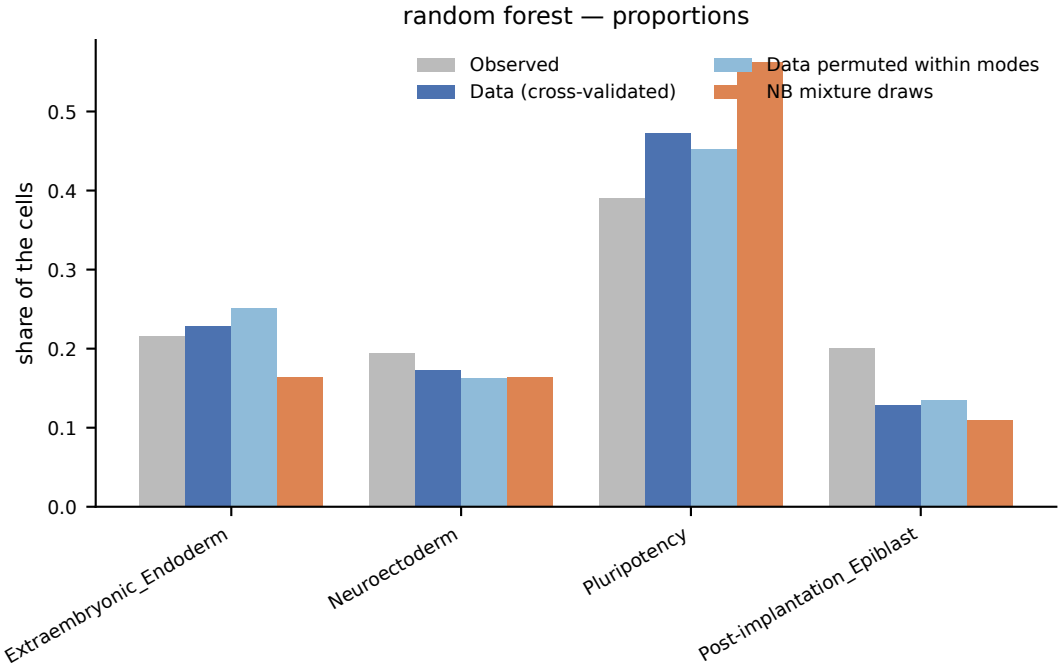
1. Generative model — quantitative checks

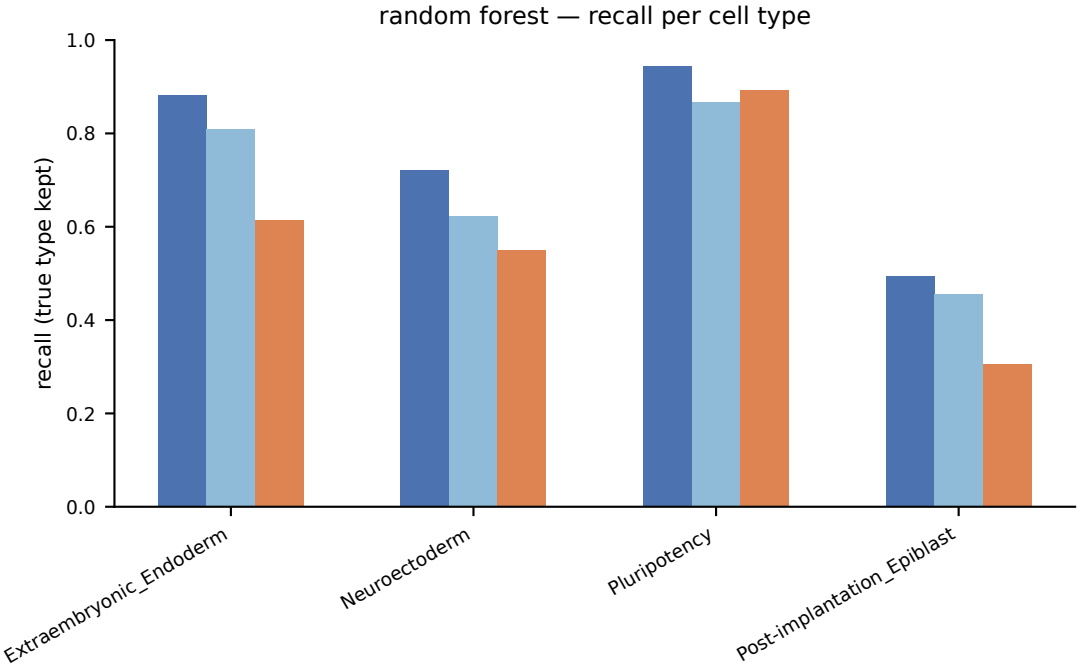
Reference = RNA of the trajectories equivalent to the simulation (growth-weighted: simulation with proliferation).

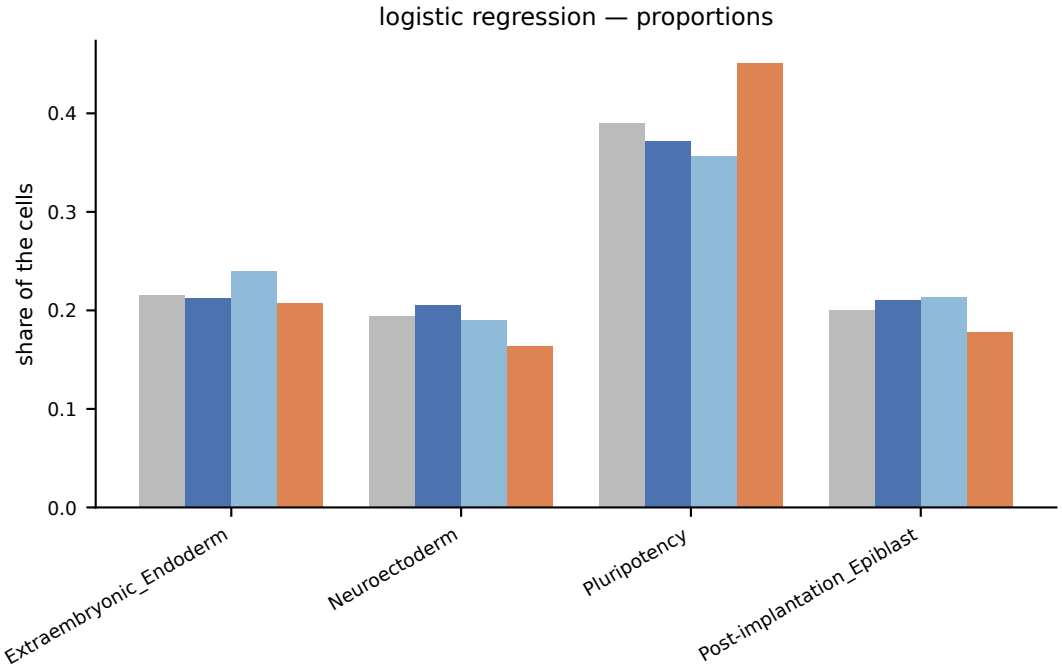


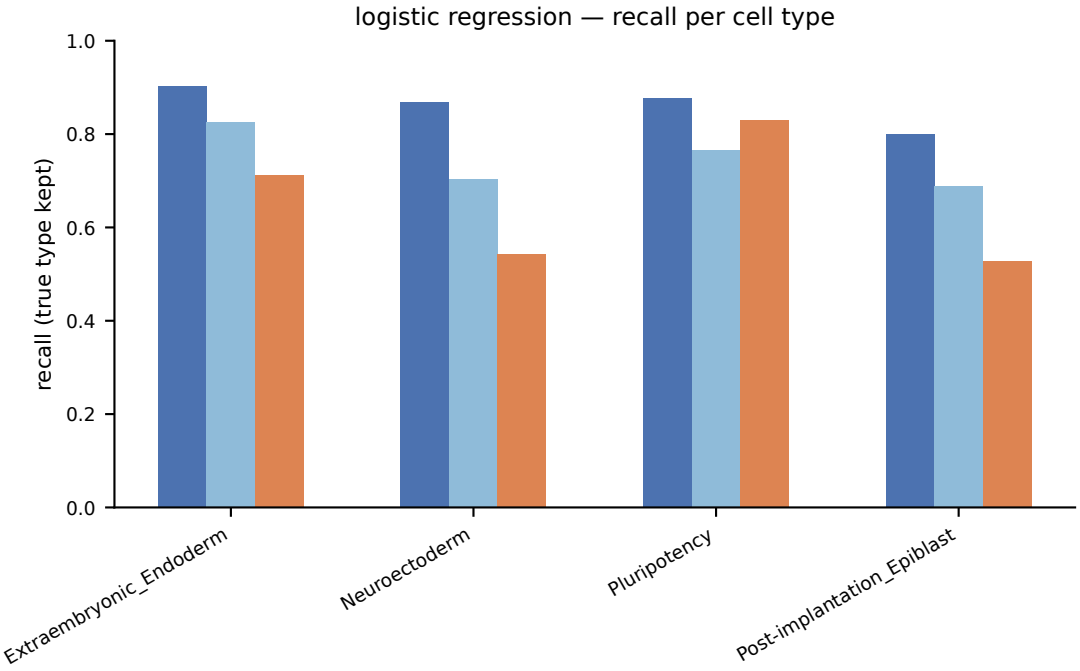
1. Generative model — cell-type identity in the NB mixture

One draw per real cell, per-sample classifiers trained on the other cells. Cross-validated data = classifier limit; permuted within modes (same sample, time, mode) = best model with these modes. Gap data → permuted: modes too coarse; permuted → draws: NB fit.







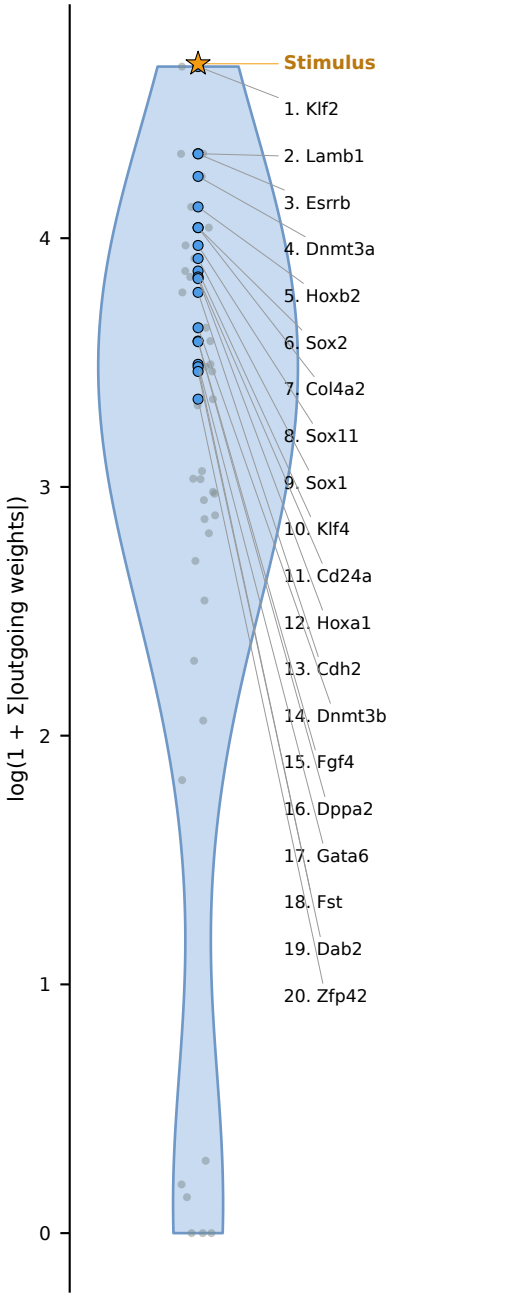


2. Gene regulatory network — regulatory power

Self-interactions excluded. Top 20 genes labelled; stimulus shown as a star (stimulus = 1.0 ≥ 0.5).

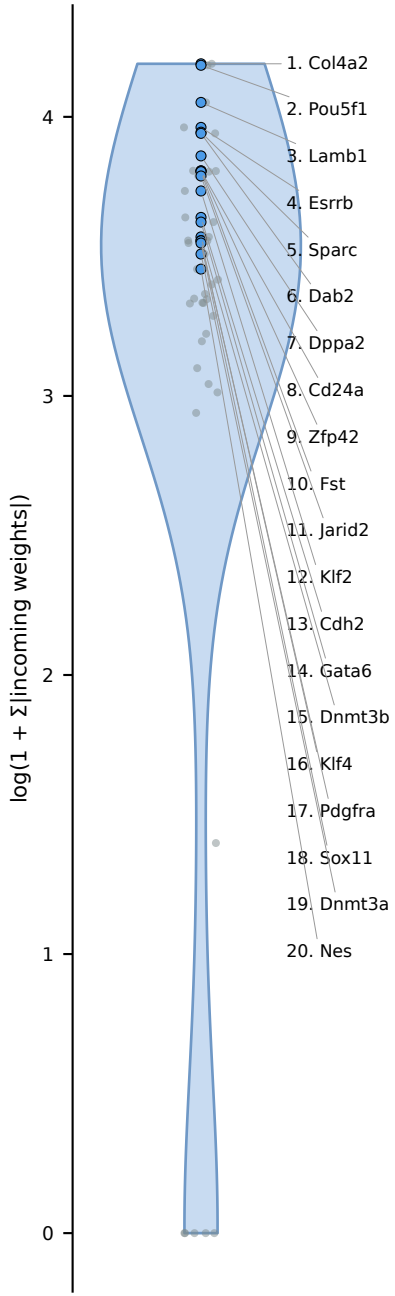
A

Outgoing regulation (regulators)



B

Incoming regulation (targets)



C

Top 20 regulators

#	Regulator	Σ|w|	act.	inh.
1	Klf2	107.85	15	20
2	Lamb1	75.69	15	5
3	Esrrb	75.64	15	19
4	Dnmt3a	69.00	19	16
5	Hoxb2	60.92	16	15
6	Sox2	55.97	18	17
7	Col4a2	55.97	9	10
8	Sox11	52.03	14	15
9	Sox1	49.32	16	12
10	Klf4	46.85	14	18
11	Cd24a	45.72	28	7
12	Hoxa1	45.42	13	16
13	Cdh2	42.91	14	8
14	Dnmt3b	37.09	20	14
15	Fgf4	35.11	18	17
16	Dppa2	35.01	20	15
17	Gata6	31.88	18	13
18	Fst	31.56	21	13
19	Dab2	30.95	12	15
20	Zfp42	27.58	21	14
★	Stimulus	109.08	23	9

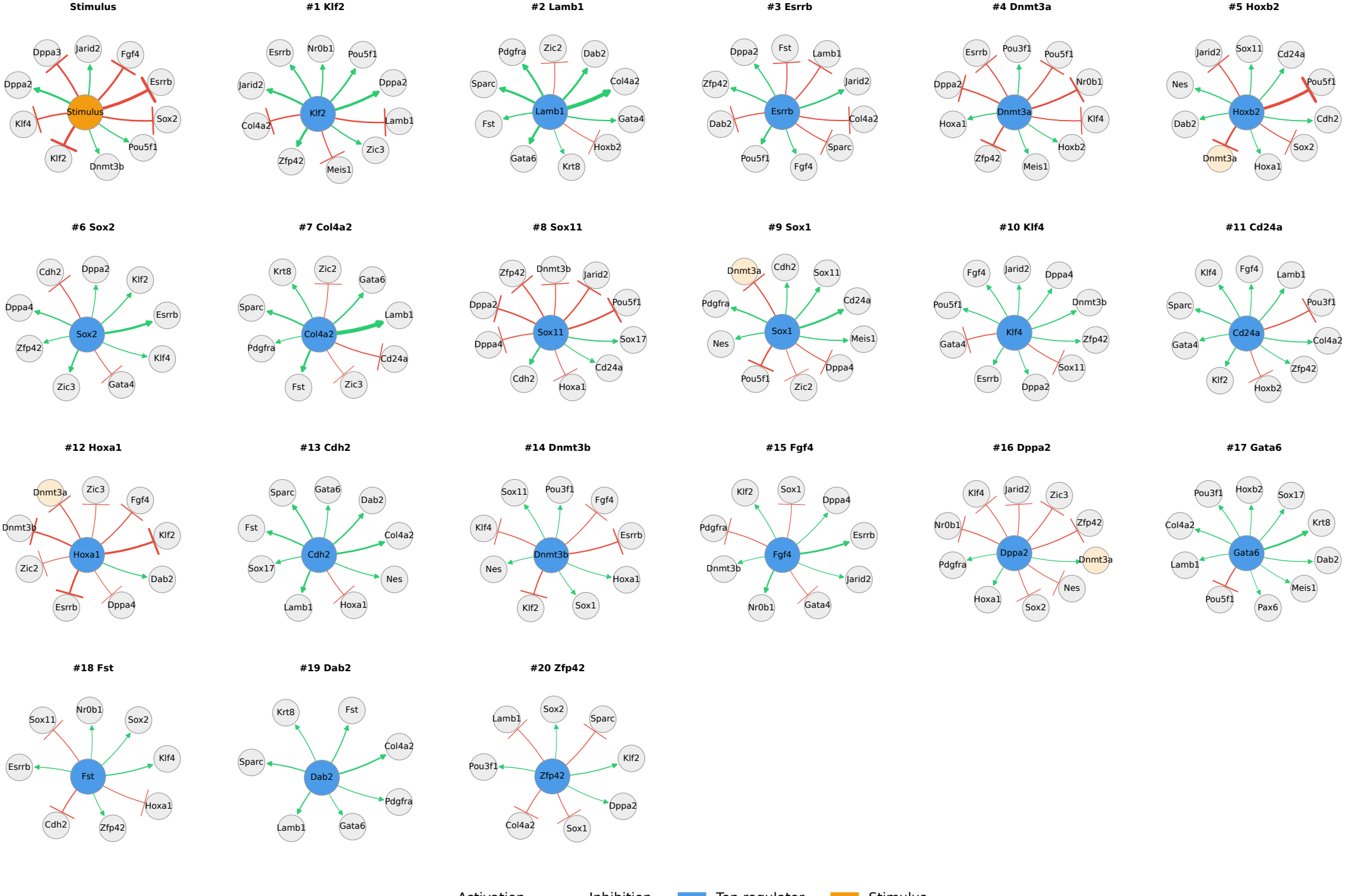
D

Top 20 regulated genes

#	Target	Σ|w|	act.	inh.
1	Col4a2	65.03	20	11
2	Pou5f1	64.68	14	14
3	Lamb1	56.50	24	6
4	Esrrb	51.57	13	8
5	Sparc	50.68	20	13
6	Dab2	50.46	23	8
7	Dppa2	46.47	14	12
8	Cd24a	44.01	17	17
9	Zfp42	43.96	15	10
10	Fst	43.86	18	12
11	Jarid2	43.17	14	13
12	Klf2	40.87	14	10
13	Cdh2	37.08	16	17
14	Gata6	36.45	20	11
15	Dnmt3b	34.51	18	16
16	Klf4	34.05	20	12
17	Pdgfra	34.02	20	10
18	Sox11	33.74	22	9
19	Dnmt3a	32.39	15	19
20	Nes	30.64	21	13

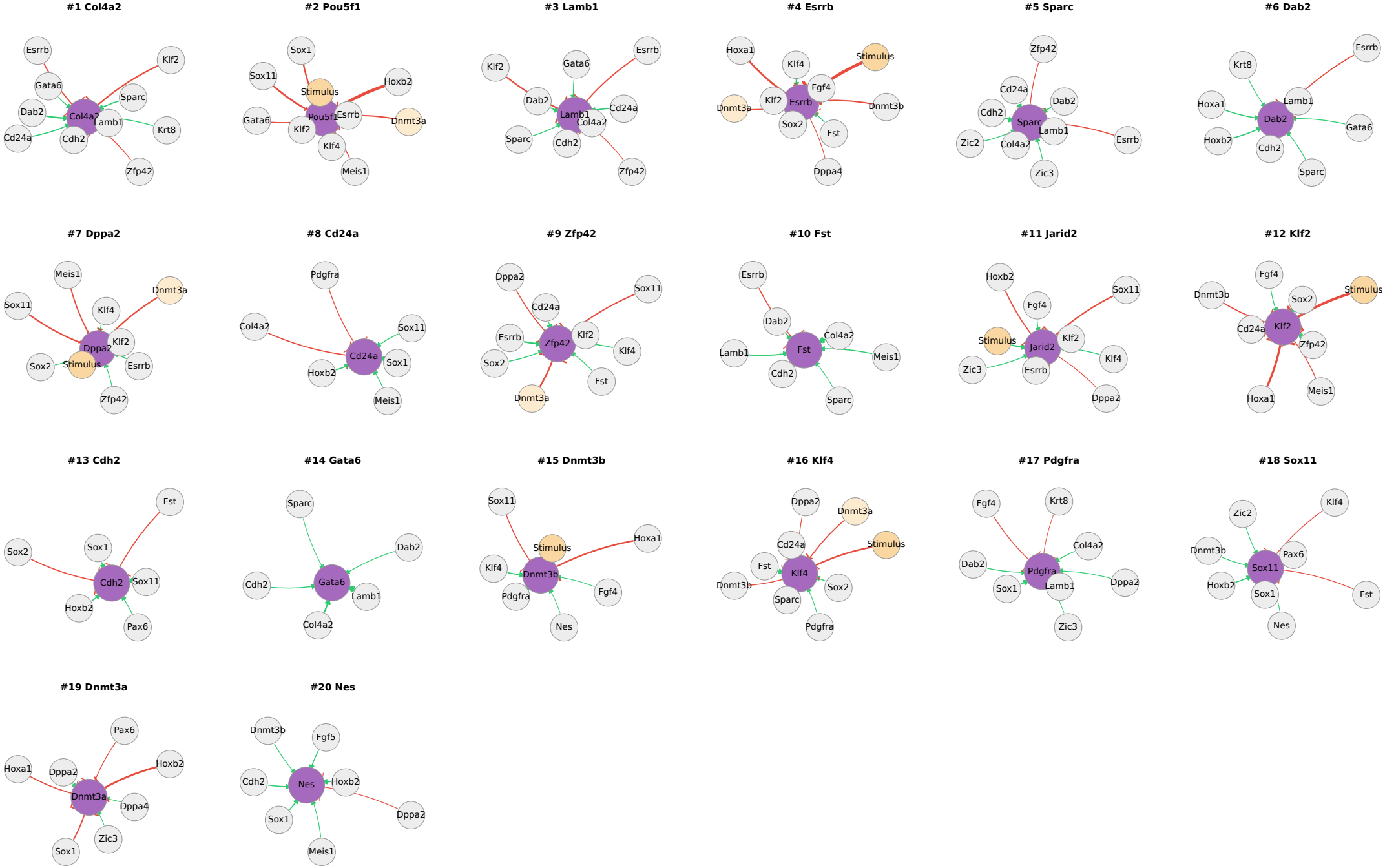
2. Gene regulatory network — top regulators and their main targets

Edges shown only if among the 10 strongest (|w|) targets of the regulator AND the 10 strongest regulators of the target; edge width ∝ |weight| (global scale).
Perturbed genes (KO/OV) are highlighted in beige.



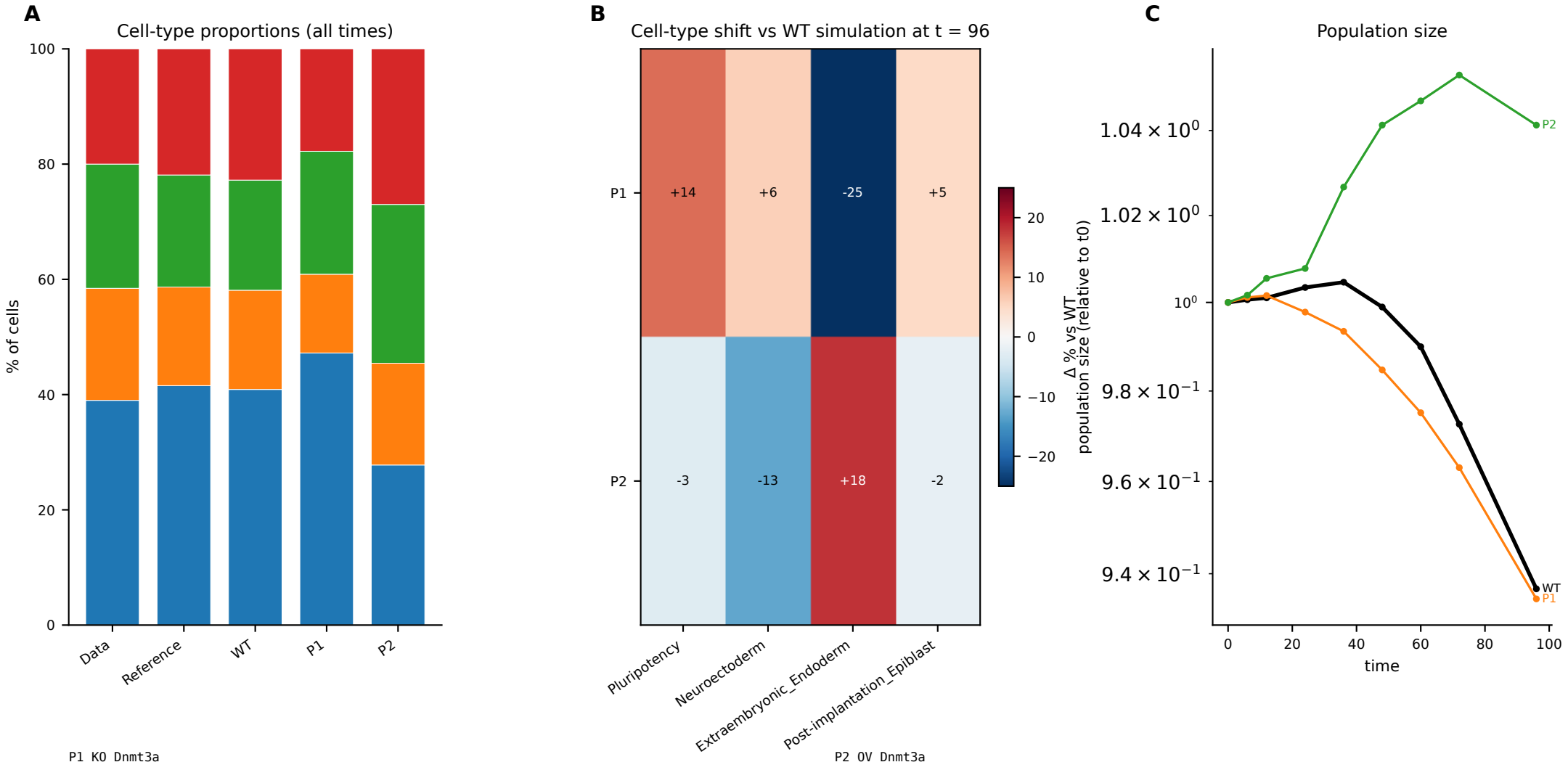
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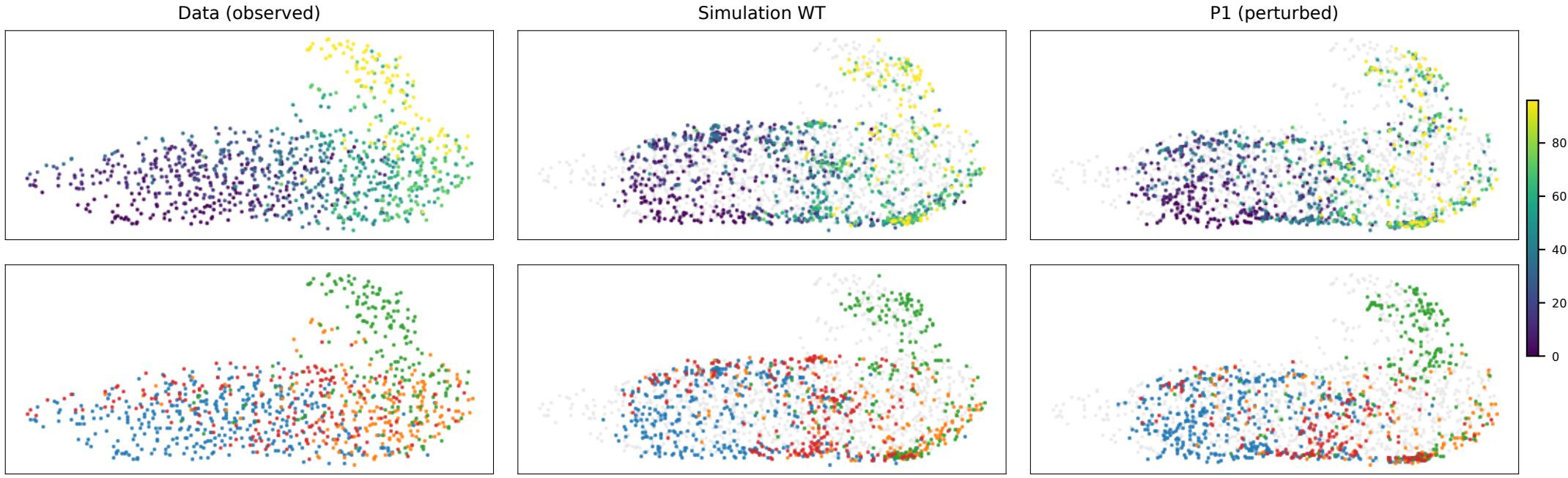
3. In-silico perturbations — overview

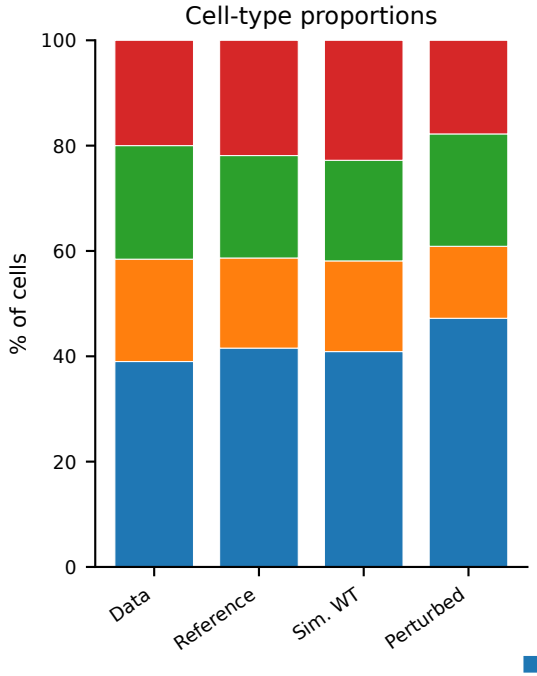
Cell types predicted by per-sample classifiers (logistic) trained on the observed data. Population sizes from the branching simulation (proliferation MLP + RATE effects).

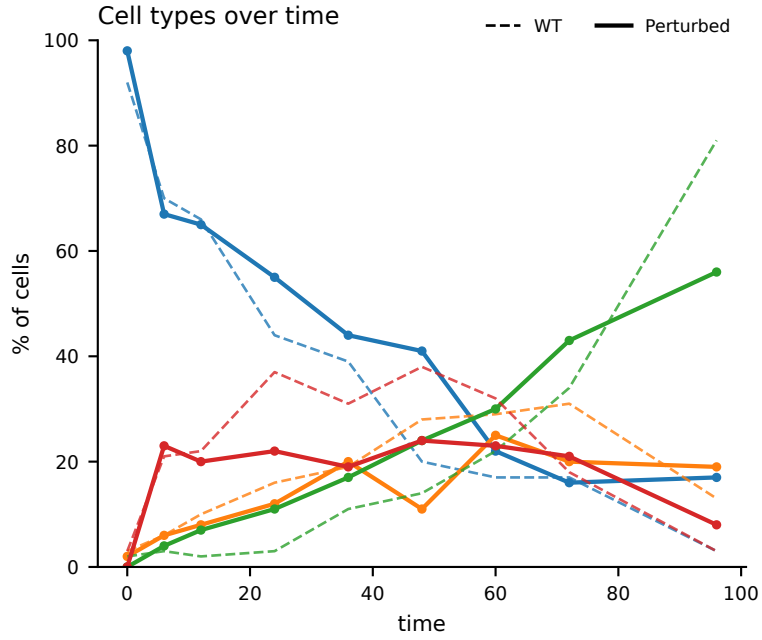


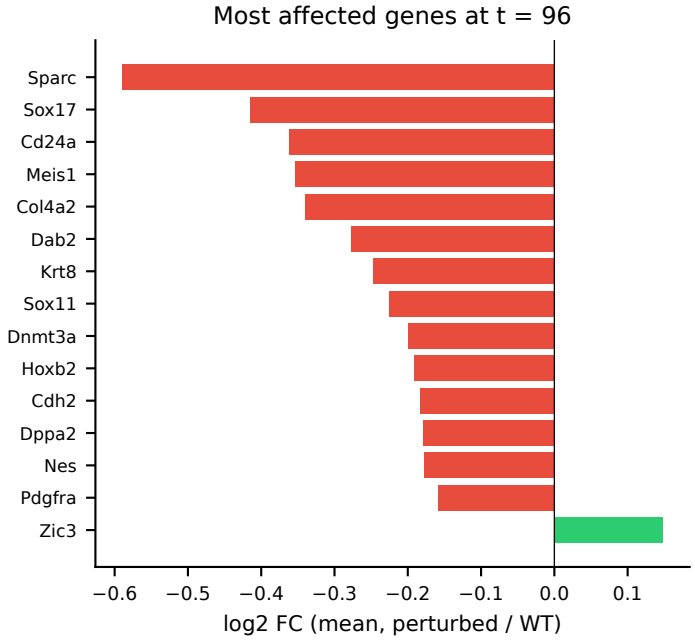
3. Perturbation P1 — KO Dnmt3a

KO_Dnmt3a_OV_none. UMAP: perturbed simulation projected on the WT embedding (grey = observed data). Population at t = 96: ×1.00 vs WT.



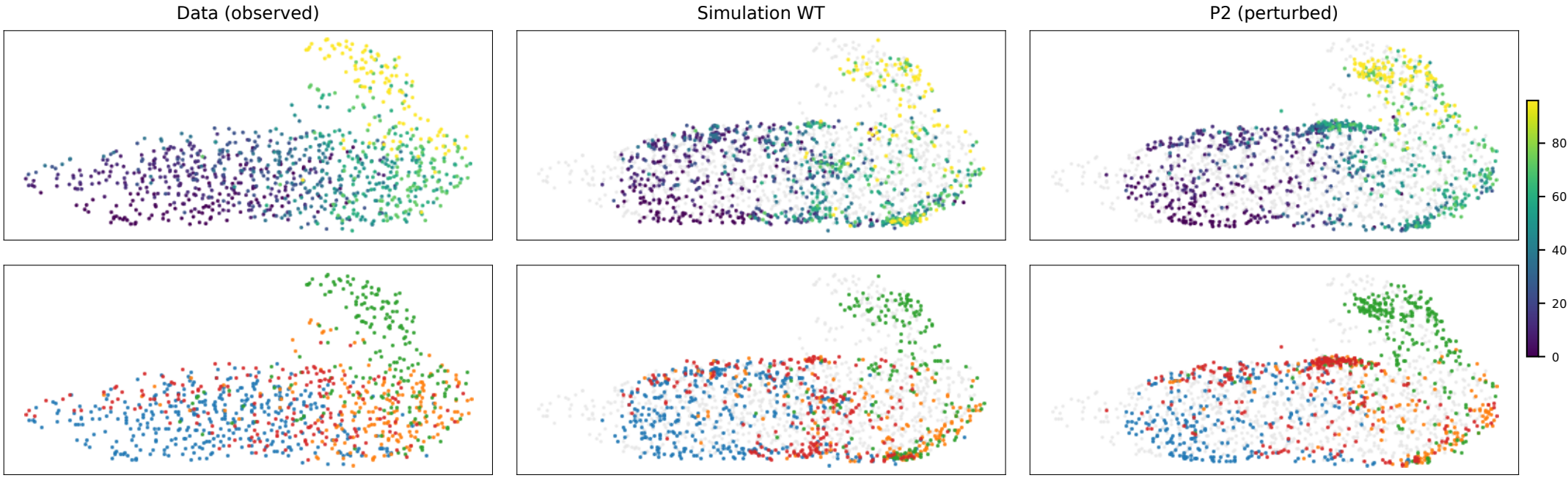


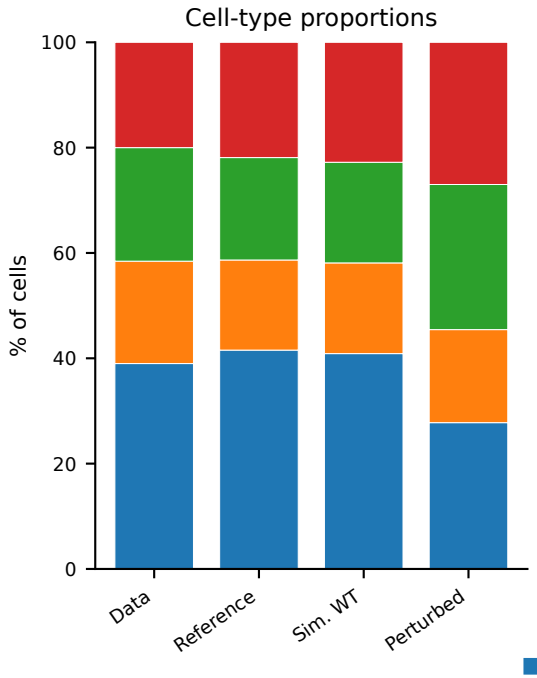


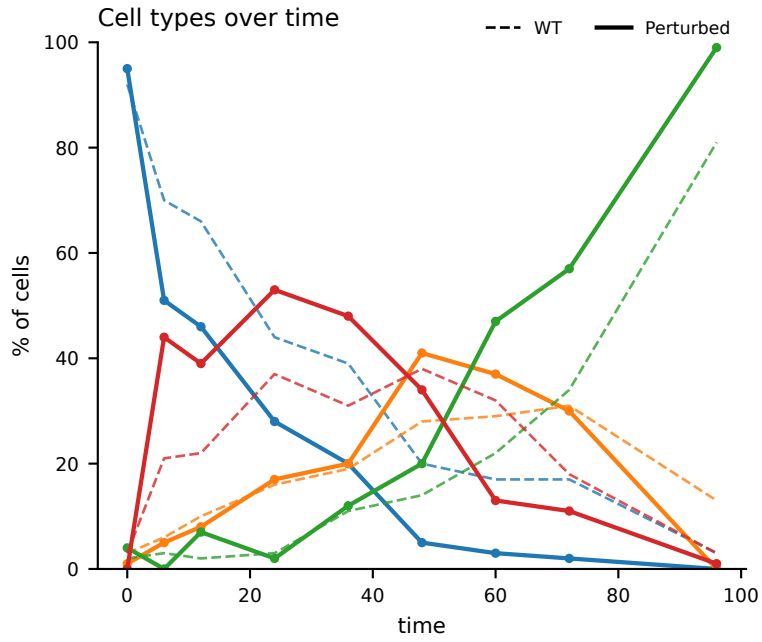


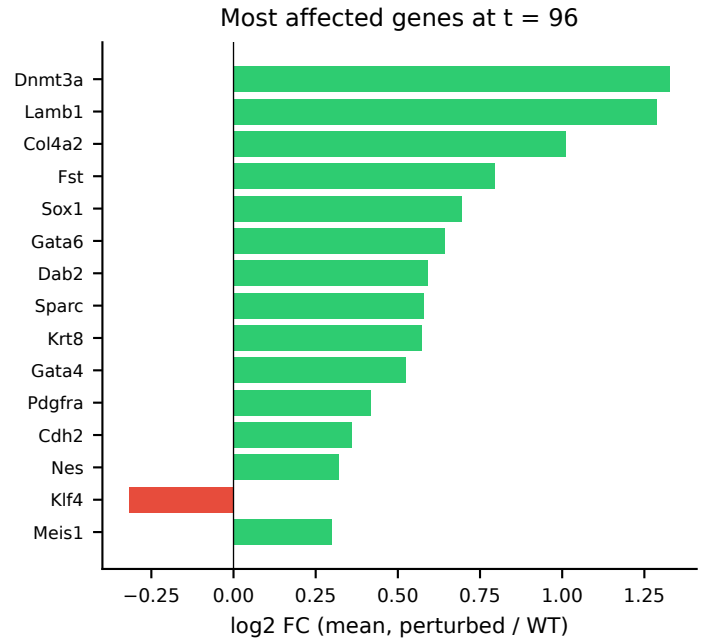
3. Perturbation P2 — OV Dnmt3a

KO_none_OV_Dnmt3a. UMAP: perturbed simulation projected on the WT embedding (grey = observed data). Population at t = 96: ×1.11 vs WT.









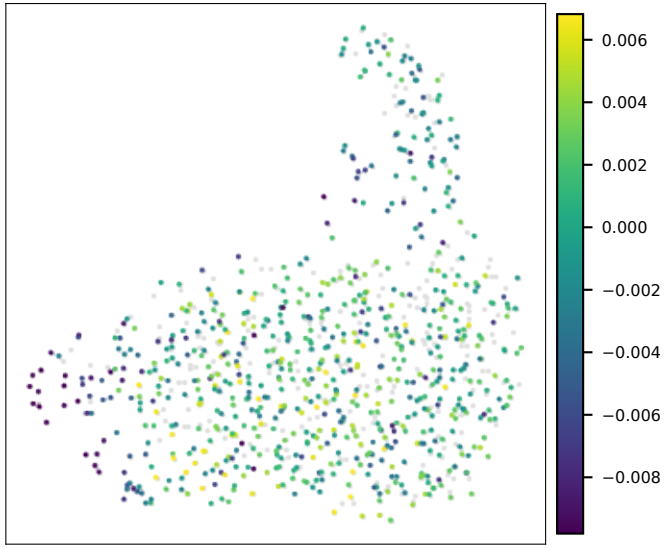
Pluripotency
Neuroectoderm
Extraembryonic_Endoderm
Post-implantation_Epiblast

4. Proliferation — prior vs learned net rate (birth − death)

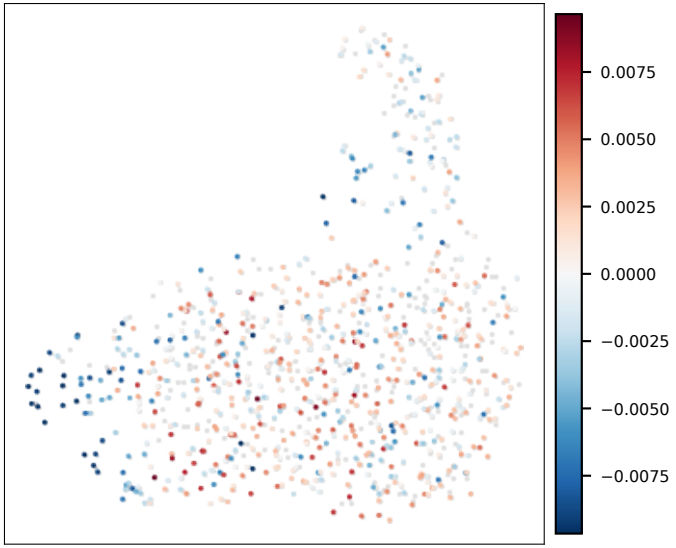
Learned = MLP R(P), mean over the trajectory states behind each observed cell (grey = cell not reached). Rates per time unit of the data.

No prior provided:
uniform net rate
(only relative rates are learned)

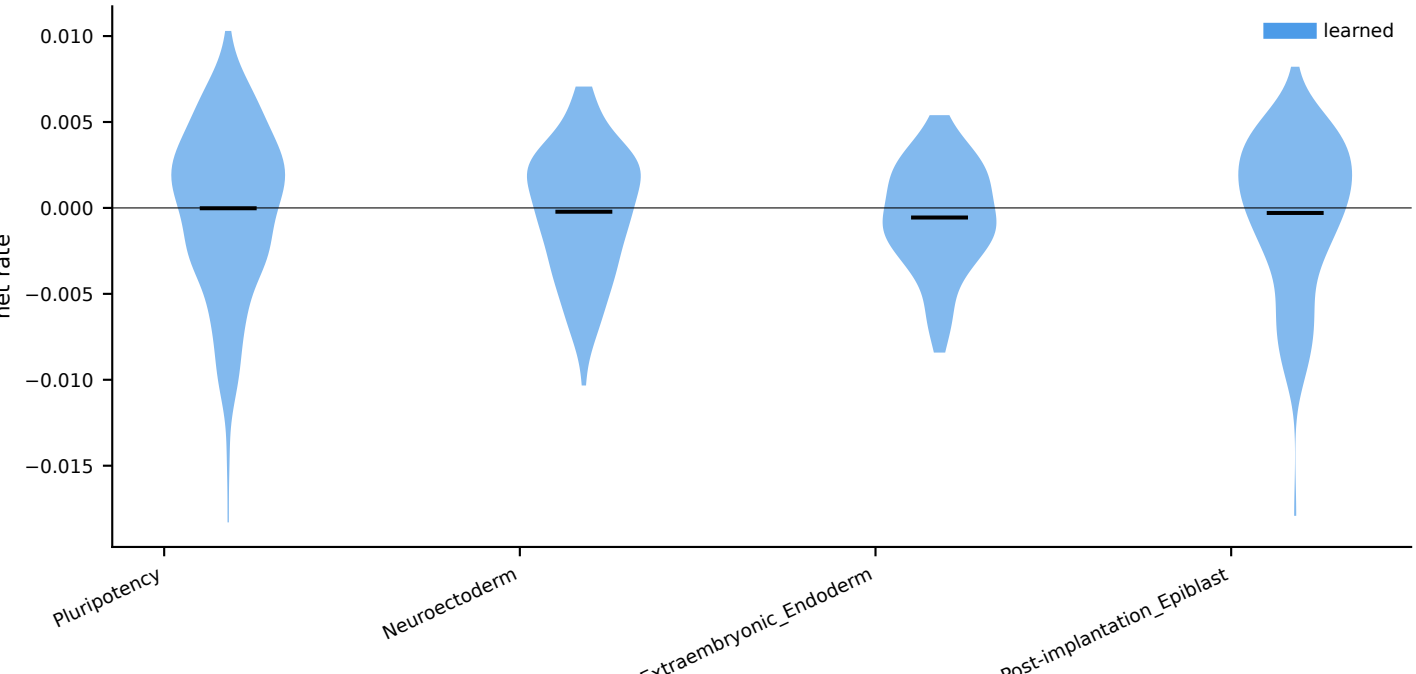
Learned net rate (MLP R(P))



Learned − mean



Net proliferation rate per cell type (cell_type)

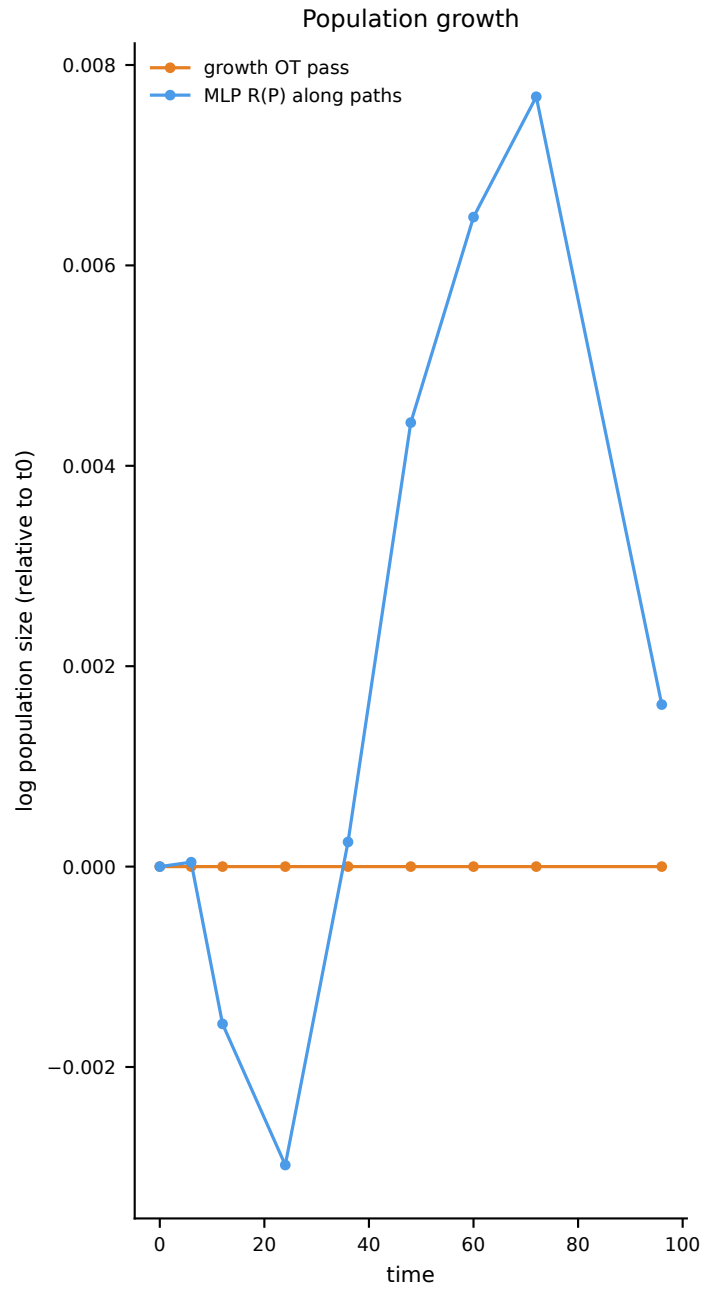


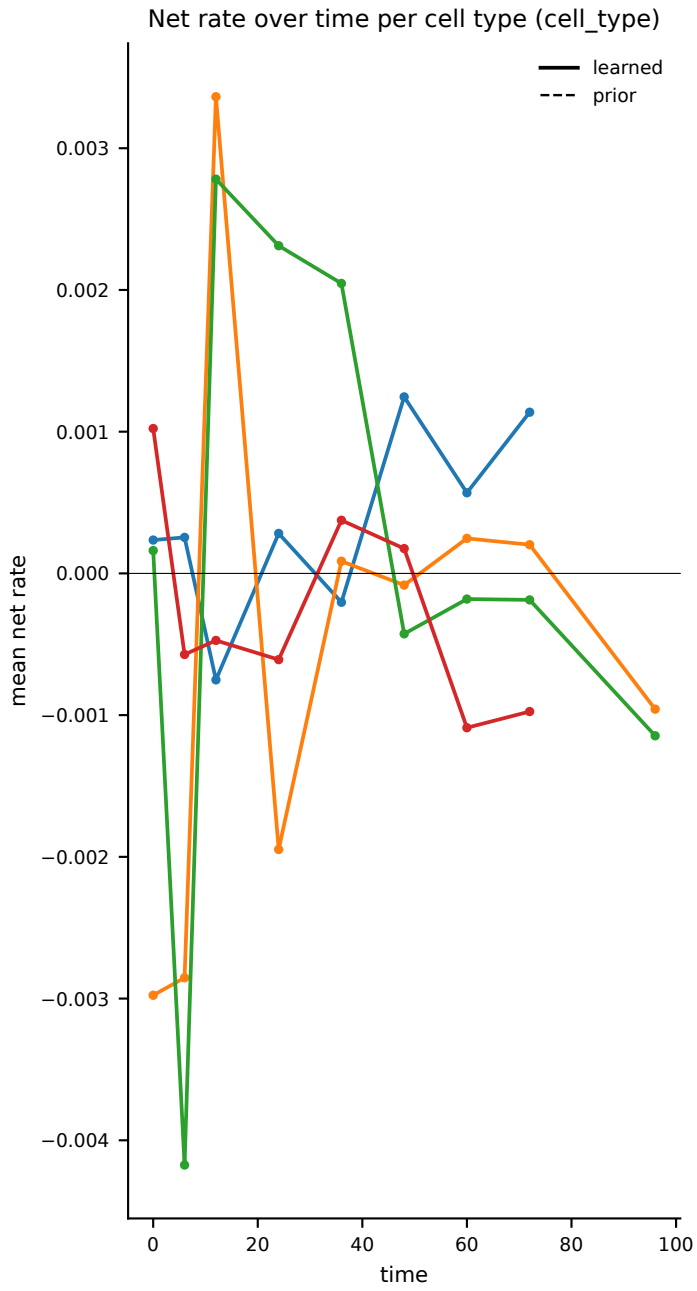
Mean net rate per cell type (cell_type)

Cell type	Anchor	Prior	Learned
Pluripotency	—	—	-1.93e-05
Neuroectoderm	—	—	-0.000221
Extraembryonic_Endode	—	—	-0.000558
Post-implantation_Epibl	—	—	-0.000293

4. Proliferation — population growth and drivers

Relative population size from the mean growth factor per interval; its absolute level follows the prior (anchored on the sheet population_sizes by fit_population_anchors, if filled).

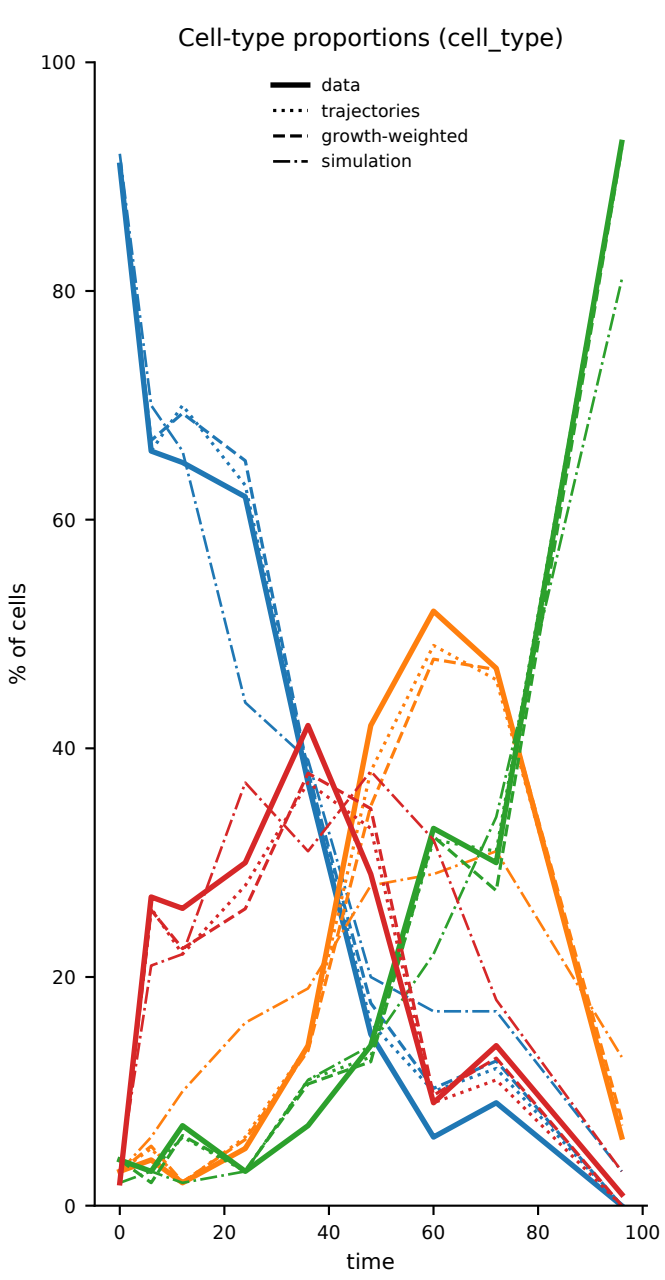


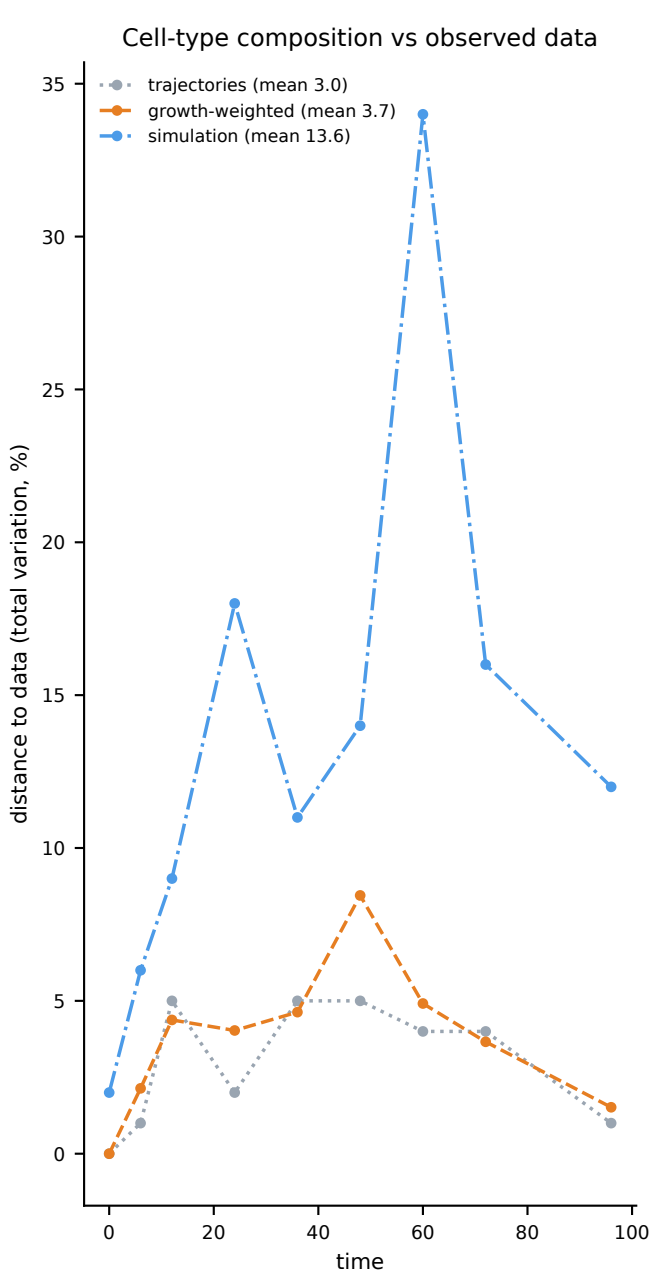


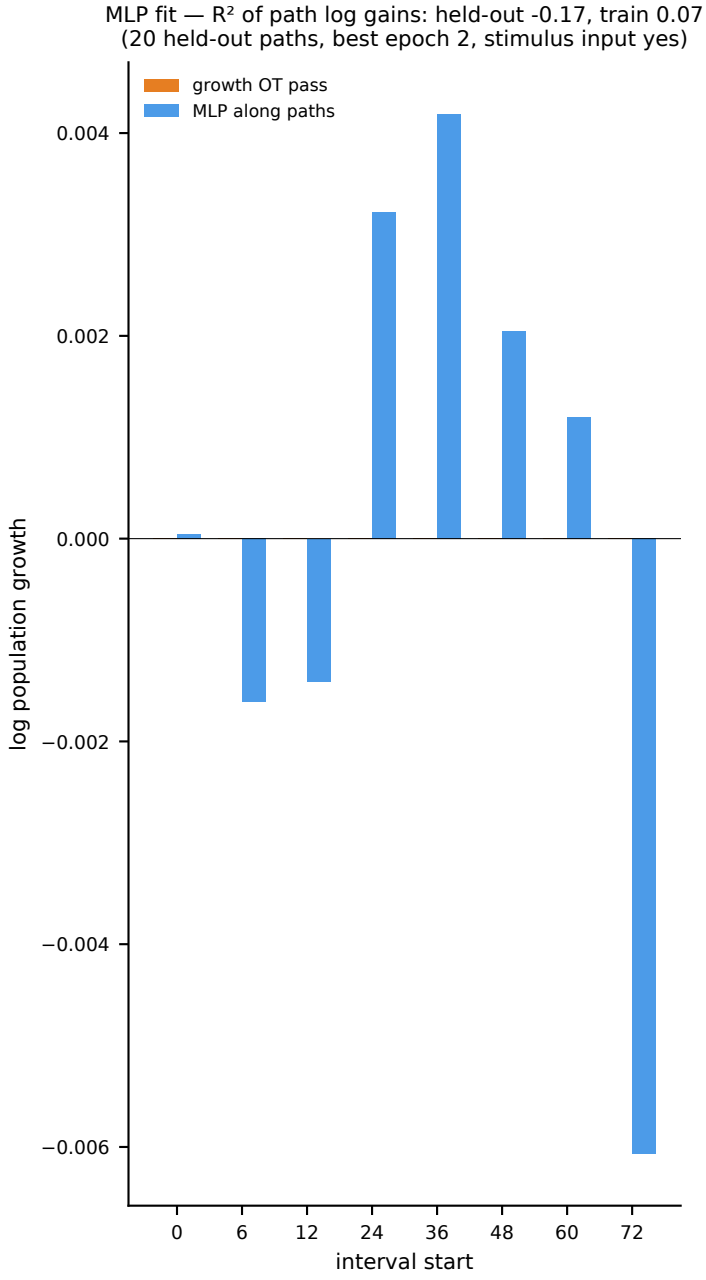


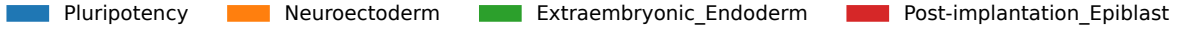
4. Proliferation — growth-weighted trajectories and MLP fit

The OT selection keeps one descendant per ancestor (trajectories without proliferation); weighting each path by its expansion exp(∫R_opt) should bring them back to the observed data. The simulation used the proliferation MLP.









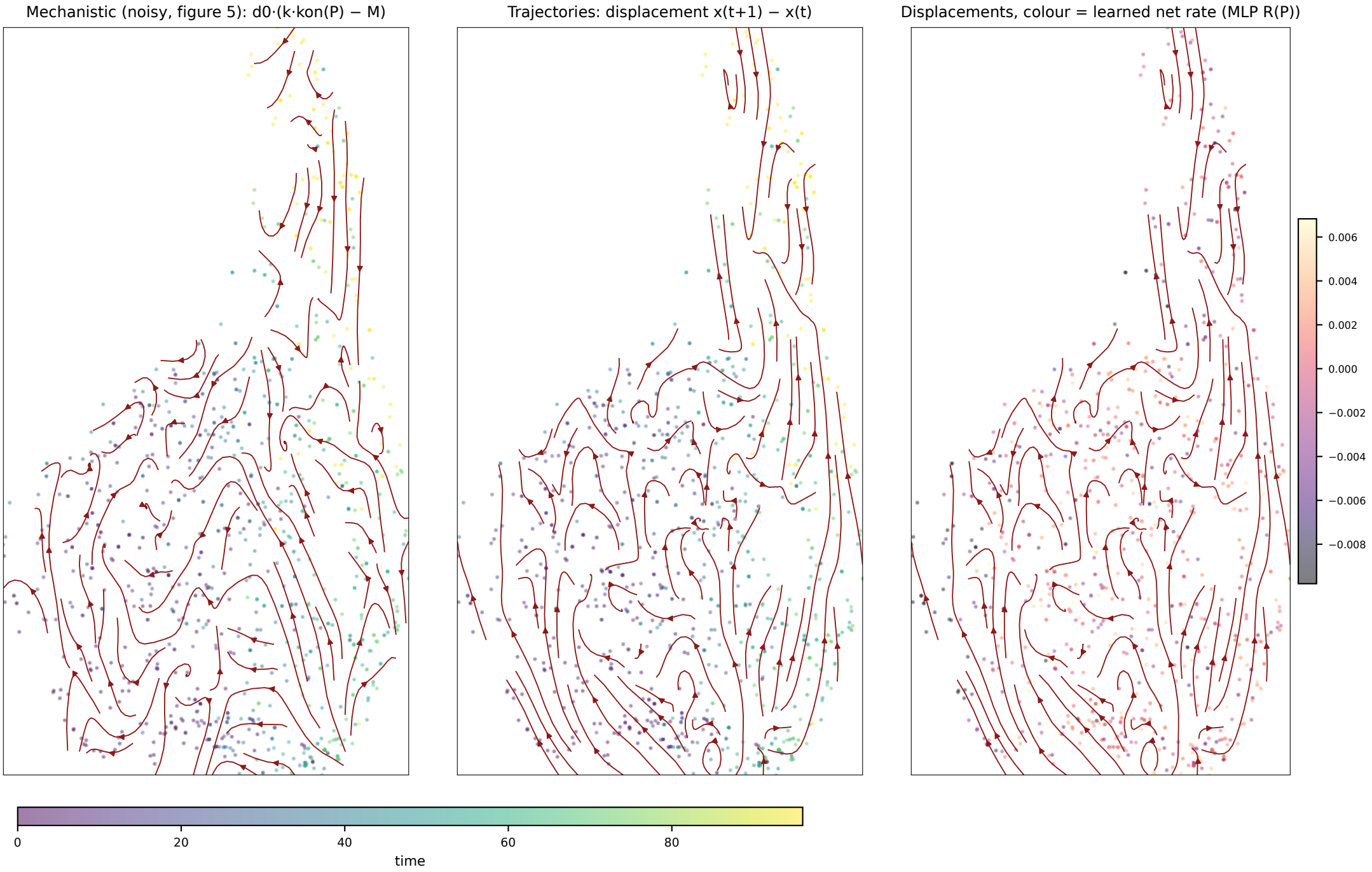
4. Protein dilution

Section unavailable:

```
ValueError('protein_dilution = False, or no birth rate (obs proliferation_birth_rate / net rate)')
```

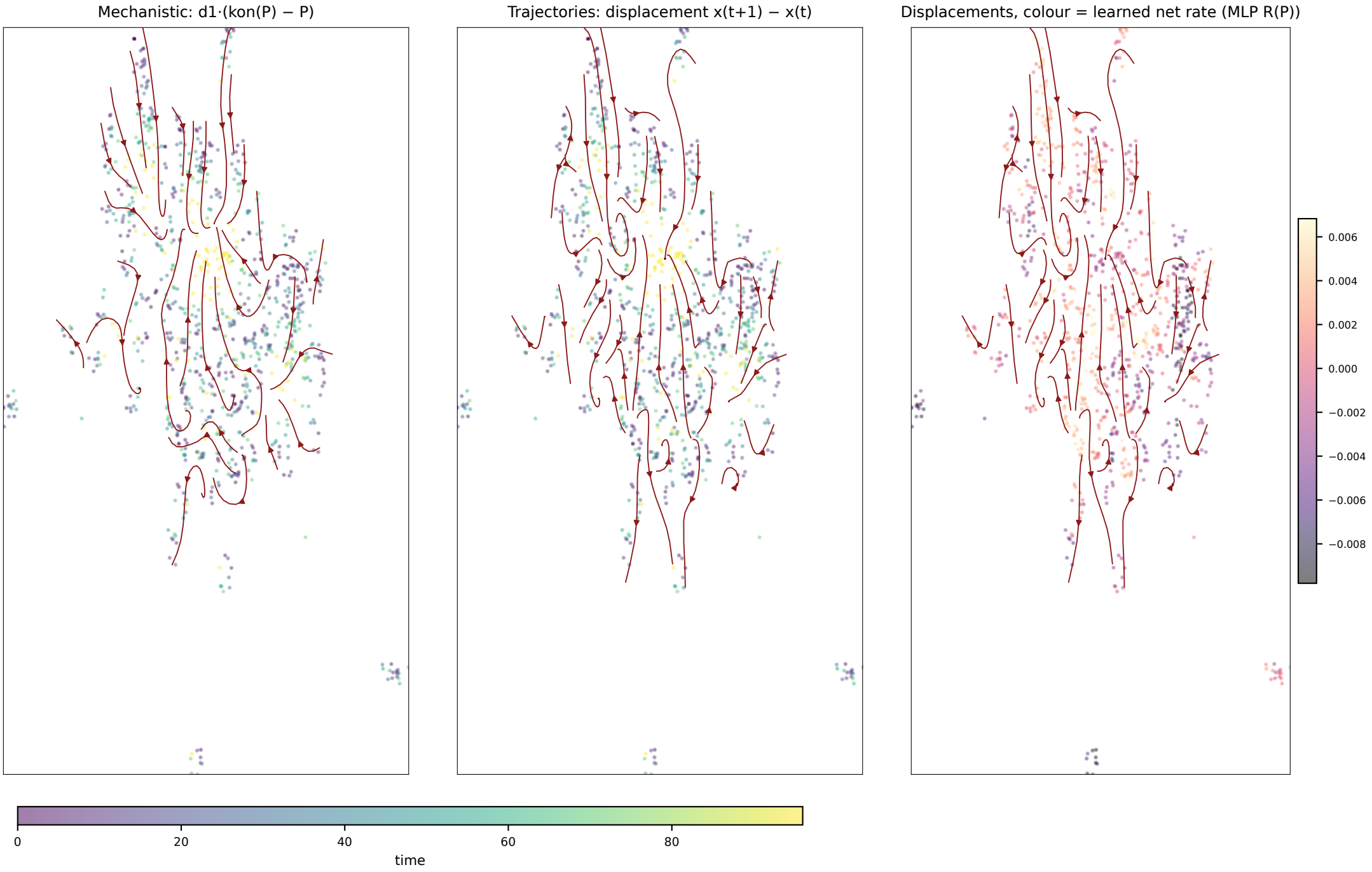
5. Learned dynamics — mRNA velocity and displacement fields

All 900 trajectory states projected onto the UMAP learned on the observed mRNA (log1p, reference depth); kNN-transition projection of the fields. Mechanistic vs trajectory velocity agreement (weighted cosine, gene space): 0.05.



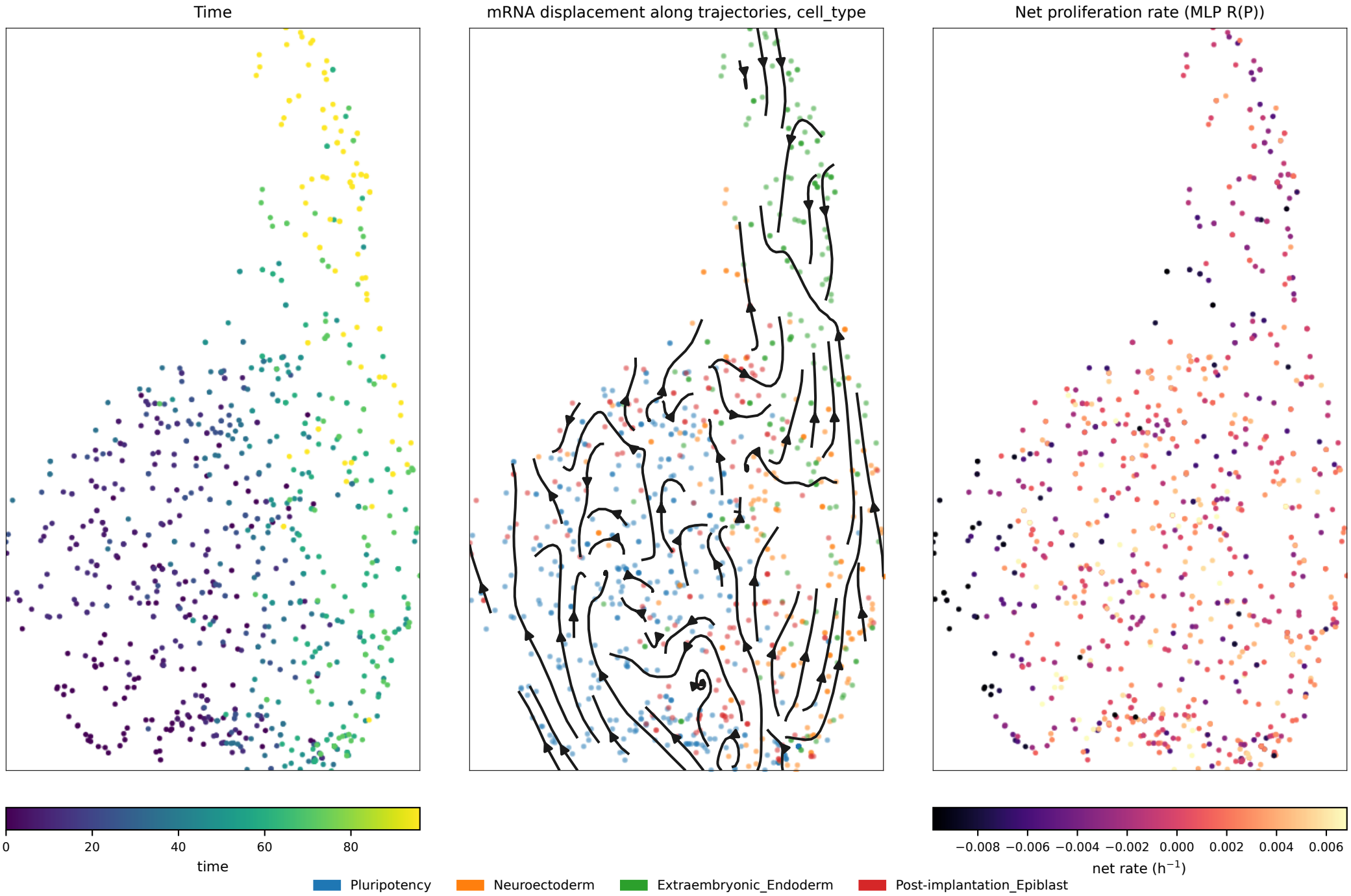
5. Learned dynamics — Proteins velocity and displacement fields

All 900 trajectory states projected onto the UMAP learned on the protein trajectories (raw levels); kNN-transition projection of the fields. Mechanistic vs trajectory velocity agreement (weighted cosine, gene space): 0.71.



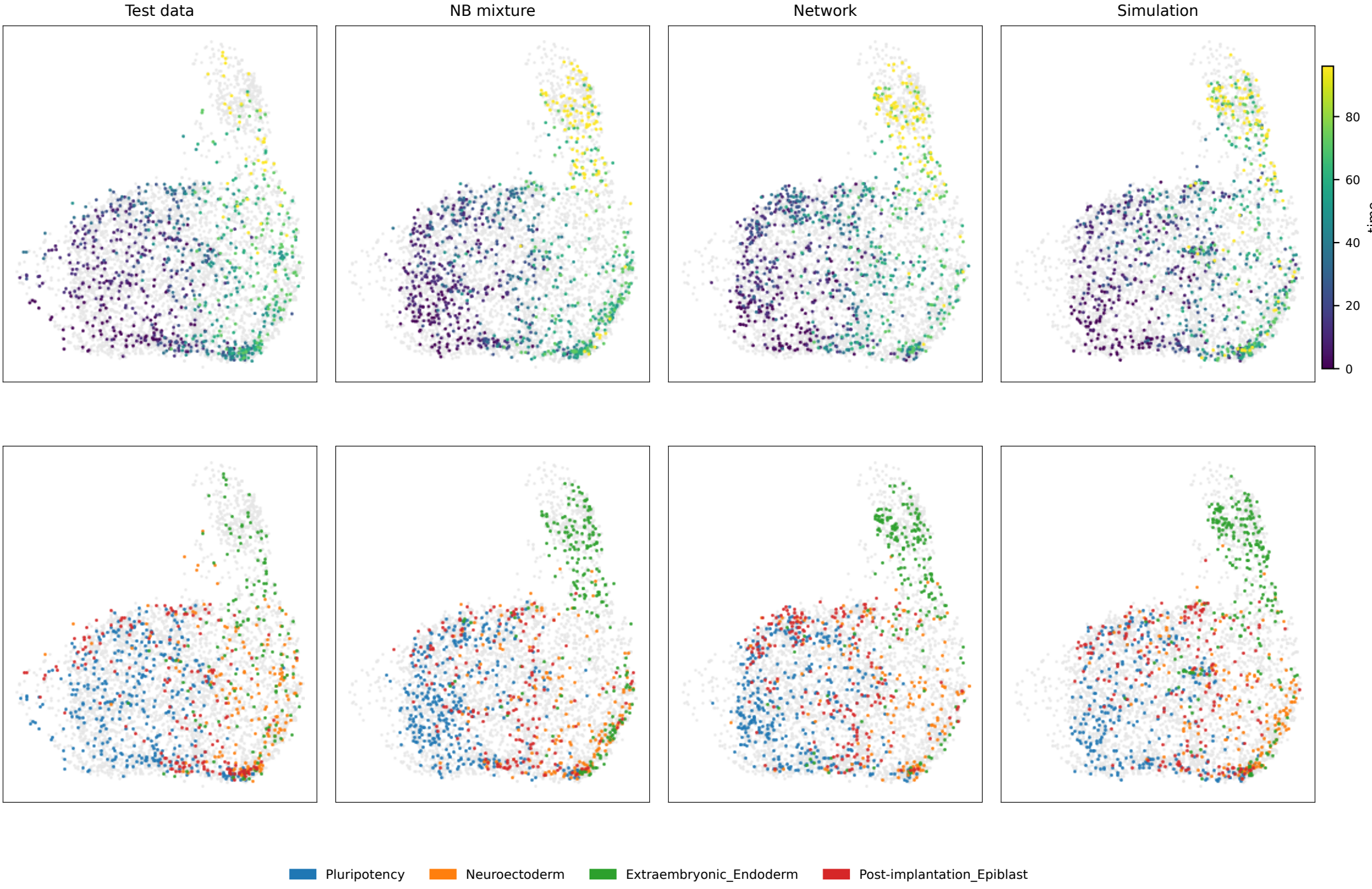
5. Learned dynamics — summary

All 900 trajectory states on the UMAP learned on the observed mRNA (log1p, reference depth); displacement along the inferred trajectories x(t+1) − x(t); background of the field: cell_type.



6. Held-out test cells — predictions with the network fixed

Test cells classified into basins with the training mixtures, trajectories inferred with the training network fixed, then simulated; projected onto the UMAP learned on the observed data (grey: training).



6. Held-out test cells — quantitative checks

Per gene: 1-D Wasserstein distance between log1p counts, averaged over the common timepoints.

